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Graduation Thesis

Benchmarking Deep Learning Models for Predicting Customer Churn in Banking with Explainable AI

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ABSTRACT

Reliable customer churn prediction is important for supporting retention decisions in retail banking. However, predictive performance alone is insufficient to establish model reliability, particularly when information leakage, probability calibration, statistical uncertainty, explanation stability, and computational cost are not adequately considered. This thesis investigates these challenges through a systematic comparison of classical machine learning and deep learning models using a publicly available banking churn dataset containing 10,000 customer records and ten predictive features.

A leakage-controlled benchmarking and validation framework is developed to evaluate five classical machine learning models (Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost) and four deep tabular models (MLP, TabNet, FT-Transformer, and RealTabR). Model development includes hyperparameter optimization, repeated stratified cross-validation, class-imbalance and architectural ablation studies, probability calibration, and threshold selection. All development decisions are completed before evaluating the models on a sealed holdout set of 2,000 customers. The evaluation further incorporates bootstrap confidence intervals, paired statistical comparisons, cross-model SHAP analysis, input-perturbation robustness, subgroup and demographic sensitivity assessments, and computational efficiency measurements.

Experimental results show that CatBoost achieves the highest holdout ROC-AUC (0.8711), while XGBoost obtains the highest PR-AUC (0.7223) and lowest Brier score (0.0986). After Holm correction, the ROC-AUC differences between XGBoost and CatBoost, FT-Transformer, and MLP are not statistically significant. These findings indicate that modern deep tabular models do not consistently outperform strong gradient-boosting methods on the investigated dataset. Cross-model explainability analysis identifies Age, NumOfProducts, and IsActiveMember as the three most influential predictors across four representative models, collectively accounting for approximately 71% of normalized SHAP importance. Additional analyses reveal that simulated missingness causes greater performance degradation than mild input perturbations and that predictive behavior varies across customer subgroups.

Based on its predictive performance and computational efficiency, XGBoost is selected for an interactive decision-support prototype. The application integrates individual churn prediction, local SHAP explanations, portfolio-level risk ranking, and model governance information. Under a 20% review-capacity scenario, the system prioritizes 400 of 2,000 customers and captures 258 of 407 observed churners, corresponding to 63.4% of all churn cases.

The findings demonstrate that reliable churn modeling requires more than selecting the algorithm with the highest predictive score. The principal contribution is an integrated evaluation framework connecting predictive performance, statistical validity, interpretability, robustness, and computational efficiency with a practical customer-prioritization prototype. Nevertheless, the use of a single public dataset without independent institutional or temporal validation limits the generalizability of the results.

The application therefore serves as a research-oriented decision-support demonstration rather than a production-ready banking system.

Keywords: Customer churn prediction; retail banking; machine learning; deep tabular learning; XGBoost; explainable artificial intelligence; SHAP; data leakage; model benchmarking; probability calibration; statistical validation; robustness; customer prioritization.

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Being able to complete this thesis work is a valuable experience of connecting the theoretical knowledge to practice of research. As a result of working on the topic of customer churn prediction in retail banking, I learned more about the difficulties of building trustworthy machine learning systems. Among the others, the study has increased my appreciation of such aspects as the proper experimental design, reproducibility, uncertainties in statistics, interpretability of models, and responsible evaluation. Moreover, implementing the application related to the work has given me the possibility to find out how the results of predictions may be converted into practical knowledge to make decisions in accordance with the limitations of the data used.

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academic as well as professional success for me. I hope that the knowledge and experience gained from this research will serve me well in the future.

All the mistakes and limitations in this thesis are entirely my own responsibility.

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LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
BCE	Binary Cross-Entropy
CI	Confidence Interval
CPU	Central Processing Unit
CRM	Customer Relationship Management
CV	Cross-Validation
DL	Deep Learning
ECE	Expected Calibration Error
F1	F1-Score
FFN	Feedforward Network
FPR	False Positive Rate
FT-Transformer	Feature Tokenizer Transformer
GBDT	Gradient-Boosted Decision Trees
GPU	Graphics Processing Unit
HPO	Hyperparameter Optimization
LR	Logistic Regression
MCC	Matthews Correlation Coefficient
ML	Machine Learning
MLP	Multilayer Perceptron
OOF	Out-of-Fold
PR-AUC	Area Under the Precision–Recall Curve
RF	Random Forest
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
RSS	Resident Set Size
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SMOTENC	Synthetic Minority Over-sampling Technique for Nominal and Continuous Features

TPE	Tree-structured Parzen Estimator
VIF	Variance Inflation Factor
XAI	Explainable Artificial Intelligence

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CHAPTER 1. INTRODUCTION

1.1 Research Background and Motivation

Customer churn, which refers to the breaking of an existing relationship with a bank on behalf of its customer, poses serious problems for retail banking. If customers decide to leave, banks might incur loss of potential revenue from services provided and lost opportunities for cross-selling and other benefits of their customer relationships. Prediction of possible customer churn could thus help prioritize how much attention and retention efforts the customer-relationship team should devote to different customers. There is evidence from previous banking research that customer characteristics can be used for statistical and machine learning models for identifying high-churn-risk customers (Mutanen et al., 2010; de Lima Lemos et al., 2022). Such predictions are more useful as evidence for examination and prioritization rather than an explanation of why customers leave.

The records of bank customers are often described by heterogeneous tables with numeric account characteristics, categorical variables, and customer engagement indicators. Supervised machine learning provides methods for estimating associations between such characteristics and an observed churn label. Linear models can establish an interpretable reference point, while tree ensembles can capture nonlinear relationships and interactions without requiring those patterns to be specified in advance. Moreover, recent advances in deep learning methods for tabular data include architectures going beyond traditional multilayer perceptron and involving representation of feature interactions through attention mechanisms and retrieval. However, empirical performance of such models is also situationally dependent. Various benchmarks on tabular data show that tree-based models optimized sufficiently well cannot be easily beaten by the neural architectures on numerous mid-sized structured datasets (Gorishniy et al., 2021; Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022).

This situation raises an interesting challenge for the use of the approach for banking analytics. The problem is not so much if a model with more sophisticated architecture performs better but if the difference in performance holds up in case the competitors were tuned sufficiently well and compared according to one and the same protocol. In a dataset where the number of observations and features is small, marginal improvement in discrimination power does not compensate the additional costs of the more complex model in terms of computation and interpretation. An informative comparison should take into account model training, scoring uncertainty estimation, probability-based ranking and its limitations.

The explainability of AI (XAI) is another factor that is worth mentioning. The predicted probability itself is unable to reveal which recorded features actually impact the prediction for an individual observation. The SHapley Additive exPlanation (SHAP) provides an example of a framework for attributing features when explaining model results (Lundberg & Lee, 2017). Nevertheless, the explanation provided by a trained algorithm depends on the specific representation of the data used in this algorithm. A

feature ranked highly in one model may be ranked differently in another one due to correlation between predictors (Aas et al., 2021).

1.2 Problem Statement

A model comparison that may seem rigorous may still result in a misleading conclusion. If the selection of any of the preprocessing parameters, synthetic observations, or threshold values was influenced by evaluation data, the scores can be too optimistic. Data leakage is a known issue in reproducible machine learning (Kapoor & Narayanan, 2023). Additionally, a single score for train-test provides very little information about variability in performance depending on sampling choices. Even if different models make predictions for the same customers, differences in ROC-AUC values do not imply superiority unless they are estimated using paired inference to consider correlations.

The definition of performance also varies depending on what is being measured. ROC-AUC evaluates the model discrimination at different thresholds, while the precision-recall curve is especially helpful if the minority class is of interest (Saito & Rehmsmeier, 2015). The probability calibration measures how well the risk predicted by a model matches its actual frequency. Threshold-dependent scores measure the outcome of the use of the decision rule at hand. A model can be good at one of them but worse at another. Resampling or reweighing classes to deal with imbalanced classes may affect recall and probability calibration differently. These distinctions motivate an evaluation that preserves the original prevalence in validation partitions and separates model training from threshold selection and final testing.

Prediction alone is not enough to ensure the responsible interpretation of results. The feature importance plots could lack stability with respect to different classes of models, manipulation of input values could change probability or decision, and aggregate measures could obscure differences between subgroups of customers. Features like gender, geography, age, and activity should be investigated in terms of their descriptive power. However, the existence of different predictions for different subgroups or for different substitutions of attributes does not mean discrimination or causality. Lastly, the issue of feasibility includes fitting, inference, explanation, storage, and memory. These aspects become particularly important when the accuracy gap between competing strong models is insignificant.

Thus, the problem considered in this thesis is a lack of integration and control of leakage for strong classical machine-learning models and modern deep tabular approaches for the same banking churn task where predictive performance should be considered along with statistical uncertainty, explanation stability, robustness, subgroup behavior, and computational efficiency. The study does not seek to introduce a new neural architecture or to claim readiness for live banking deployment. It seeks to establish what can be concluded reliably from a controlled benchmark and how those conclusions can support a bounded decision-support demonstration.

1.3 Research Objectives and Research Questions

The general aim is to contrast classical machine learning models and deep tabular learning models for predicting customer churn at banks in a reproducible and leakage-free research setup, as well as to assess the relevance of their results.

Specific aims are to (a) create common data, predictors and division into development and hold-out samples for nine models; (b) study the influence of tuning, class imbalance handling, probability calibration, and fixed classification thresholds; (c) assess uncertainty and compare models based on paired statistical tests; (d) check if there are consistent important predictors and their ranks in representative architectures; and (e) check robustness, sub-group behaviour, demographic inputs sensitivity and computational performance. The obtained results are used to show a limited proof of concept of ranking and explaining individual churn predictions without any extra prediction experiment.

The four research questions that guide the empirical study are:

- **RQ1.** How do classical machine learning techniques and deep tabular methods perform comparatively when assessing discrimination, probability quality, and locked threshold classification performance under the same leakage-controlled development and evaluation process?
- **RQ2.** Is there statistical evidence for the differences observed in performance? How does the handling of class imbalance, probability calibration, and threshold selection affect the performance evaluation?
- **RQ3.** What are the most influential predictors of churn and their ranking by importance among competing boosting, multi-layer neural network, and transformers architectures?
- **RQ4.** What is the robustness of the models to imperfect data inputs, subgroup performance, direct sensitivity to demographic variables, and computational efficiency?

The four research questions cover similar areas but address different elements of model robustness. RQ1 is about the predictive outputs that have been generated through measurement; RQ2 pertains to the strength of the comparison and development decisions; RQ3 addresses how reliably competitive architecture explains those predictive outputs; while RQ4 explores whether the generated models are informative despite being constrained by data quality and resources.

1.4 Research Scope and Boundaries

The empirical analysis is based on the public dataset Churn for Bank Customers, which includes 10,000 records of customers and 2,037 labels for churning (20.37%), with ten predictors: CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. Target variable, Exited, indicates whether the customer left the bank or not. RowNumber, CustomerId, and Surname variables are not used for prediction; CustomerId is used only for aligning records. The dataset does not have information on customers' complaints, their level of satisfaction, transaction histories, previous contacts for retention, and observation time.

Consequently, results should be considered associations within a static public benchmark rather than proven early warning ability of the model on a bank's future customers' stream.

The nine models selected for evaluation include five classical algorithms (Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost), as well as four deep tabular methods (multilayer perceptron, TabNet, FT-Transformer, and RealTabR). Splitting the dataset into development (8,000 records) and holdout (2,000 records) sets is done only once via stratified sampling. Hyperparameter tuning, resampling ablations, early stopping, calibration selection, and decision-threshold selection are conducted only on development set. Repeated stratified cross-validation supports development-stage comparison; the final fitted models are evaluated on the fixed holdout without subsequent changes to the locked prediction outputs. Statistical uncertainty is then assessed from these immutable outputs rather than by revising model settings in response to test performance.

Model interpretation is examined in the greatest depth for XGBoost, CatBoost, MLP, and FT-Transformer by means of a common permutation-SHAP test protocol. An additional TabNet explanation test acts as an additional agreement metric. Similar explanations cannot be claimed to be found for the original final RealTabR model since the attempted reconstruction did not fulfill the prediction-fidelity tolerance requirement. Perturbed inputs, subgroup measurements, and sensitivity analysis characterize the model behavior for the stated testing conditions. The memory test consists of two post-test measures on the frozen setup and should not be considered a measurement of the original fitted models.

The accompanying application prototype shows how an already chosen XGBoost model is capable of individual risk estimation, model explanations, and constrained review list construction. The governance dashboard demonstrates the relevant facts taken from the benchmark. This prototype employs the same input ten features and research artifacts that were used in the benchmark. This is a prototype for possible decision support system and not an independent institutional validation or automatic retention process. Live customer data, intervention results, retention cost measurements, external banking databases, and production load tests are not included. Accordingly, neither the experimental benchmark nor the application establishes external generalizability, causal drivers of churn, legal fairness, or operational deployment readiness.

1.5 Research Contributions

First, there is the establishment of a carefully designed benchmark in which the recently developed deep table architectures compete against a set of strong conventional and gradient boosting baselines. Each of the candidates is optimized for its respective hyperparameters and subsequently compared using the same repeated stratified cross-validation procedure applied to development data. The goal is to protect conclusions on the usefulness of deep learning from a reliance on a poorly performing comparator of classical approaches.

Second, there is a comprehensive protocol for predictive validation and statistical comparison. Development and hold-out data are distinguished, imbalance and pre-processing are included within the proper training partitions, calibration methods and thresholding are chosen before any testing, and hold-out predictions are left immutable. Discrimination, probability calibration and threshold-locked performance are examined as separate aspects. Confidence intervals and paired comparisons using bootstrapping and multiple comparisons correction are employed to establish whether differences between nominal rankings are statistically significant.

Third, an empirical study of explanation agreement. Instead of considering one visualization of SHAP as the ground truth, the research compares feature rankings from two different gradient-boosting models, one feedforward neural network model, and one transformer model in the same feature space under the same explanation process. Rank correlation, top features' consistency, resampling stability, and feature direction consistency measure which interpretation patterns are consistent in representative machine learning architectures. The additional TabNet comparison is done for testing the agreement of interpretation patterns, while the problem of fidelity of RealTabR is openly mentioned rather than replaced with interpretation by a different estimator.

Fourth, an assessment of reliability and feasibility in a wider perspective. Perturbations of numerical and categorical features and simulation of missing data assess sensitivity to imperfect data. Subgroup comparisons and evaluations of geographic and gender dependence assess differences not visible through aggregation of scores. Training and prediction costs, model complexity, explanation cost, and memory consumption estimate the relation between predictive performance and computational resources. Such analysis is diagnostic: it neither proves causality nor guarantees fairness and readiness for application.

The prototype application allows the empirical study to have an applied context. The XGBoost model, which was archived, is utilized for the generation of risk scores and explanation and the prioritization of a small number of customers for further human inspection. The combination of presenting prediction for each customer, and for a portfolio of customers as a whole with evidence of the selection and governance process demonstrates how the output of the scientific work can be presented to practitioners. This demonstration is not considered as an additional prediction validation effort and does not imply that having high estimated churn probability suggests a good intervention strategy.

1.6 Thesis Organization

The remaining five chapters range from the underlying concepts of the study to the experiment and its limitations. Chapter 2 describes bank customers' churn prediction, classical and deep learning approaches to tabular classification, the criteria for evaluation, explainable AI, and empirical literature in this field. Chapter 3 outlines the dataset, the experimental setup that prevents data leakage, model architectures, validation procedures, statistics used, the explainability process, robustness testing, and reproducibility controls.

Chapter 4 presents the results of the empirical study, which include repeated cross-validation, ablation analysis, calibration, locked hold-out performance, statistical testing, agreement of the explanation, robustness, subgroups' characteristics, and computational resources. Chapter 5 discusses the findings in relation to the research questions and existing literature, the application prototype based on XGBoost and the workflow of customer prioritization, implications for practice, and validity concerns. Chapter 6 offers brief answers to the research questions and summarizes the demonstrated contributions, as well as external validation and future research priorities.

CHAPTER 2. THEORETICAL BACKGROUND AND RELATED WORK

2.1 Customer Churn Prediction in Retail Banking

2.1.1 Churn as a predictive and decision-support problem

Customer churn implies the ending or cessation of the relationship of a customer with a service provider. Customer loss in the context of retail banking decreases the worth of potential accounts, loans, payments, and investments and requires additional customer acquisition. Prediction is needed due to the fact that retention specialists cannot pay the same attention to each relationship. The function of the model includes the calculation of the probability or ranking to understand the customers that require further investigation. In other words, the model is different from the intervention strategy since prediction only reveals the likelihood of the problem while not indicating what measures can be taken to solve it (Mutanen et al., 2010; de Lima Lemos et al., 2022).

Supervised learning framework consists of rows having customer information and a binary output. Specifically, the model will analyze the relationship between the variables and the target variable to forecast the score of the unknown observation. This approach is useful for customer-specific banking data as the model learns about demographics, account holder status, financial condition, and engagement. Nevertheless, the score cannot be viewed as the cause of the departure from the bank. Observational variables could just serve as proxy variables of unobservable processes while the model trained on historical label does not have to evaluate the impact of a particular retention treatment.

In addition, the operational definition will rely on the name of the record and the method of observation. If the record is labeled as one for an exited customer, then an observation event will be evident in the data set that, on its own, does not define the prediction period, but rather does not contain the time-stamped events leading up to the churn. The conclusions that we get from the data table above will act as a benchmark for prediction cross-sectionally. These cannot be called validation of early warning predictions at all.

2.1.2 Characteristics of tabular banking data

The financial features are, in most cases, heterogeneous: the age, balance, and rating are numeric, the geographical location and gender are categorical, and the product ownership is a small discrete variable. All these factors may influence the model's predictions through thresholds, nonlinear connections, and interactions. For example, the dependency between product ownership and attrition does not need to be linearly dependent on the number of products. Moreover, since various models have different feature representations, the preprocessing stage is the necessary part of the modeling and cannot be considered to be neutral for all candidates.

The middle range structured data problem creates its own methodology choices. Growing model complexity allows capturing interactions missed by linear classifiers, while more optimization and higher sensitivity to evaluation strategy become necessary. Empirical research on tableaux demonstrates that gradient boosting trees continue to be

powerful competitors for neural network solutions in particular, especially in cases of limited sample size and heterogeneous input variables (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022). The correct question therefore is not which model type is the absolute winner, but which one works well in similar conditions.

2.2 Classical Machine Learning for Tabular Classification

2.2.1 Logistic Regression and Random Forest

Logistic Regression estimates the log-odds of a binary response variable as an additive function of the input variables. To derive a probability from 0 to 1, it applies a logistic transformation to this linear combination of predictors. This technique is computationally efficient, can use regularization, and provides a benchmark regarding whether the given data contain enough signals to be modeled with an additive decision boundary. It is not that Logistic Regression cannot classify non-linear responses in principle; the problem is that non-linearities and interactions must be accounted for via proper feature engineering.

A Random Forest is an ensemble of decision trees which is generated from the same set of predictors using randomly selected training datasets with random subsets of the variables. Through splitting criteria, each individual tree is able to incorporate non-linear and interaction effects automatically without the need to define any functionals. Ensemble of trees reduces sensitivity of this approach in comparison with an individual tree, even though probability and decision thresholds of the forest may need to be calculated separately. Random Forest has been developed by Breiman (2001). In the work of de Lima Lemos et al. (2022), who examined banking customer churn, a Random Forest outperformed a number of traditional approaches.

2.2.2 Gradient-boosted decision trees

Gradient boosting works by constructing the sequential additive ensemble of trees where later trees correct shortcomings of the fitted model at the moment. Nevertheless, XGBoost is the implementation that takes into account regularization and computational optimizations to build the trees effectively and support sparse inputs (Chen & Guestrin, 2016). The principle behind the work of LightGBM consists in the usage of histograms and is connected to gradient-based one-side sampling and exclusive feature bundling that increases the computational speed in some cases (Ke et al., 2017). Finally, CatBoost implements ordered boosting and categorical features handling to prevent the target encoding-based prediction shift (Prokhorenkova et al., 2018).

Tree-based models split the feature space and are able to represent prediction discontinuities and interactions between numerical and categorical features. The ability to do so is important when working with banking data because the combination of those features like the customer's engagement status, age, and other characteristics of accounts may carry some information that additive logistic regression cannot capture. However, in some cases, the boosting algorithms are not necessarily better. The choice of parameters like tree depth, number of trees, learning rate, subsampling and

regularization affects both fit and overfitting; wrong tuning might result in overfitting or wasting resources.

It is impossible to use an ineffective linear model to prove the effectiveness of a deep network. For neural networks to beat current approaches in prediction of tabular churn, they have to be compared to gradient-boosted models that already proved themselves on structured data. This hypothesis is grounded on benchmarking results and the requirements of model optimization (Gorishniy et al., 2021; Grinsztajn et al., 2022). More details about the tuning strategies, available pre-processing procedures and the final configuration will be described in Chapter 3.

2.3 Deep Learning for Tabular Data

2.3.1 Multilayer Perceptron and TabNet

A Multilayer Perceptron (MLP) maps a numerical feature encoding via fully connected layers and activation functions. As compared with a linear classifier, it is capable of learning interactions without having to design all cross-products explicitly. The effectiveness of an MLP is influenced by its architecture dimensions (width and depth), feature scaling, optimization, regularization, and stopping criterion. Dropout decreases dependence on particular activations of hidden units during training, while normalization techniques influence optimization dynamics; none of these components is necessarily useful without empirical investigation. Modern tabular datasets indicate that the quality of MLP training and default configurations can make a significant difference in comparisons with tree ensembles (Holzmüller et al., 2024).

TabNet involves a sequential attention-based feature selection scheme. In every decision process step, the mask generated by the model guides it toward a certain portion of inputs to which a sequence of representations can contribute to the final prediction (Arik & Pfister, 2021). This approach is meant to allow the model to utilize its capacity in an intelligent way and provides for architecture-specific explanation of feature usage. However, TabNet's internal masks cannot be interpreted the same way as SHAP values generated from prediction probabilities. A common external explanation protocol remains useful when explanation results must be compared across unlike architectures.

2.3.2 FT-Transformer and retrieval-augmented learning

FT-Transformer considers each input feature as a token and applies transformers-style attention to learn representations and feature interactions. Firstly, numerical and categorical variables are represented in a compatible way as tokens and only then information is integrated via attention layers. FT-Transformer was proposed by Gorishniy et al. (2021) in the course of a systematic reevaluation of neural architectures for tabular data representation. While attention may help to enhance expressivity, it still entails certain architectural and computational choices. Consequently, the capability of a transformer to learn interactions has to be evaluated empirically on the target dataset and not vice versa.

Retrieval-augmented tabular learning augments representation of a target record with relevant information from a set of reference records. TabR consists of a feedforward network and a nearest neighbor-like retrieval module that provides retrieval contexts and enhances predictions based on them (Gorishniy et al., 2024). The RealTabR estimator considered in the current study belongs to the family of the approach under the PyTabKit framework. Thus, it is considered as the implementation variant of retrieval-augmented tabular learning and not as a new architecture proposed in the current research.

2.3.3 What tabular benchmarks establish and what they do not

Empirical work does not provide evidence for a consistent deep-versus-tree hierarchy. Gorishniy et al. (2021) discovered strong sensitivity to model choice and benchmark, Shwartz-Ziv and Armon (2022) stressed the practical power of XGBoost, and Grinsztajn et al. (2022) highlighted persistent benefits of tree models in many traditional tabular tasks. However, Holzmüller et al. (2024) were able to reduce this gap for some datasets using better MLP implementations and good default choices. Both results are compatible: on the one hand, old claims about automatic deep learning superiority were overestimated; on the other hand, neural baselines cannot be deliberately underoptimized.

For a churn benchmark, thus, model comparison should clearly distinguish the power of an architecture from the advantage of an experimental setup. Tuning regime, early stopping, scaling, treatment of categoricals, and data access may play an independent role regardless of model type. An appropriate experiment uses an adequate model-specific pipeline for all candidates while keeping prediction target, amount of training information, evaluation data, and reporting rules unchanged. The current research follows this definition of model comparison; the specific controls are detailed in Chapter 3.

2.4 Reliable Model Evaluation

2.4.1 Leakage prevention, validation, and imbalance

The issue of information leakage stems from information not available at the intended time for predictions that affects the development of a model or its evaluation. Information leakage might be even present in models where the target column is omitted: scaling prior to splitting, feature selection based on evaluation labels, and creation of oversampled samples in training and validation splits might bias estimates towards an overoptimistic performance. Kapoor and Narayanan (2023) note that leakage poses a significant risk for reproducibility in science employing machine learning techniques. The proper isolation thus should include not just the final model but all stages of preprocessing, selection, training, calibration, and evaluation of a model.

Development-holdout approach leaves out a collection of observations that will be assessed without any bias related to the possible choices that can affect the model. Stratified cross-validation can be done in the development datasets where the distribution of outcomes in the folds is preserved and, by repeating cross-validation, we

can reduce our dependence on the one particular random partition. Holdout will be done after completing the process of model construction rather than discussing possible ways to improve score repeatedly. It ensures that the final assessment will serve as a test for the established predictor even though it does not provide any time or cross-validation itself.

Class imbalance becomes another element of distortion. If the number of churners is low, a classifier that classifies almost all the data into the majority class can yield high classification accuracy but will lack any practical value in predicting churn. Class balancing involves changing the training criterion while oversampling means changing the distribution of the samples from the minority class. SMOTE makes use of interpolation of the minority class samples to generate additional training samples (Chawla et al., 2002); if categorical features are also present in addition to the numeric variables, SMOTENC should be used. Oversampling is limited to the training split while the true distribution should be maintained in the validation and holdout sets.

2.4.2 Discrimination, probability quality, and decisions

Predictive performance can be seen in at least three ways. Discrimination is about whether positive instances get higher scores than negative ones. ROC-AUC measures ranking throughout all the potential decision thresholds, whereas the precision-recall approach takes the positive class into account even more and is important where the positive outcomes are rarer. Its baseline precision is determined by churn frequency; PR-AUC scores must be interpreted considering the distribution of the target (Saito & Rehmsmeier, 2015). Neither of these metrics implies the fact that the 0.70 score implies 70% actual event occurrence.

Calibration is about the numerical interpretation of predicted probabilities. Calibrated model makes predictions where the probabilities correspond to the actual outcomes in the right-sized groups. The Brier score evaluates the squared probability error and is good for both calibrated and discriminating predictions, while log loss punishes overconfident predictions most of all. Expected calibration error reflects the error of probability bins depending on the binning strategy used. Studies of classical and neural classifiers show that good discrimination does not ensure good probability estimates and that post-hoc sigmoid or isotonic adjustments can be useful, although improvements are not automatic (Niculescu-Mizil & Caruana, 2005; Guo et al., 2017).

However, threshold-dependent classification has its own purpose. Precision reflects the percentage of customers identified as potential churners who indeed leave, recall reflects the percentage of actual churners detected, while specificity reflects the percentage of true positives of non-churners. F1 score takes into account both precision and recall, while balanced accuracy takes into account sensitivity and specificity. Matthews correlation coefficient takes into consideration all four elements of the confusion matrix and is valuable for comparison when there are different numbers of classes (Chicco & Jurman, 2020). Threshold setting implies choosing between competing criteria and must be done based on the development dataset only.

Capacity-constrained prioritization is connected with classification but cannot be confused with classification. A bank having the capacity to check the 20% of customers with the highest scores may perform top-k ranking even when the list of these customers does not coincide with the group of predicted positive under fixed threshold of classification. Recall and lift at chosen capacity serve as descriptive metrics of ranking under resource constraints. Without information about intervention outcomes and costs, no conclusion on incrementality can be drawn.

2.4.3 Statistical uncertainty and paired comparison

Performance estimations differ since a finite set of observations provides only one realization of the customer population. Bootstrap resampling may be used to quantify variability about a statistic through recomputation of this statistic using the resamples (Efron, 1979). If models are assessed on the same customers, the samples have to be paired in order to quantify correlation in the estimates of models' performance differences. Overlapping confidence intervals for the models do not substitute paired testing of the difference between these models.

The DeLong test procedure compares correlated estimates of ROC-AUC, whereas paired bootstrap comparison allows assessing differences between the metrics for which an easy analytical test is hard to construct (DeLong et al., 1988). The McNemar test is designed to find the discrepancies between paired binary outcomes and thus deals with the estimand different from the difference between discrimination. In case there are multiple pairs of the models and outcomes tested, uncorrected p-values result in false discoveries. The Holm procedure controls family-wise error rate in multiple testing (Holm, 1979). It is important to understand that failure to reject the hypothesis of equal performances does not mean that these performances are equal.

2.5 Explainable AI and Model Reliability

2.5.1 SHAP, feature dependence, and explanation stability

The goal of explainable artificial intelligence is to ensure that the behavior of predictive systems is more interpretable. In terms of the Shapley value theory, SHAP provides the feature attributions in relation to the base prediction of a predictive model (Lundberg & Lee, 2017). In the local setting, SHAP shows how feature values of a particular data point shift the model prediction from its base prediction. The aggregation of SHAP attributions for all records in a cohort results in global feature importance ordering. They differ in the sense that the former is a measure of typical importance in the explanation cohort, while the latter describes a particular record.

A SHAP explanation relies on the choice of the reference distribution and missing features representation. The correlated features may split the attributed importance, and some perturbation approaches could create feature groups which are not seen in the real joint distribution of the features. The dependence problem has been considered by Aas et al. (2021). To make a fair comparison of the two machine learning architectures (boosting and neural networks), it is needed to compare them by an architecture-independent measure and within the same feature space. Nevertheless, attribution needs

to be seen as the characteristic of the model/explanation and cannot be interpreted as the causal effect of customer behavior.

Interpretation based on a single model may be erroneous despite the impressive plots. Consensus among the models may be calculated based on the Spearman or Kendall rank correlation measures as well as overlapping of the leading features, whereas re-sampling of the explanation data by bootstrap may help to detect whether the leading predictors are the same across all samples. Consensus does not guarantee that the selected feature causes the churn, but it definitely increases the confidence that the rankings are not unique for this model only. It is critical to verify the fidelity of the explanation when it was obtained from the restored model.

2.5.2 Robustness, subgroup behavior, and responsible use

This kind of estimate based on a clean test table will not necessarily preserve its stability when faced with faulty data entry or data incompleteness. Input perturbations, input categorization mistakes, and input values missing can affect the probability estimation, ordering, and thresholding in different ways. Input perturbation helps us understand those weaknesses because we can manipulate the input according to some rule while keeping the function estimate constant. However, this does not give us proof that the model will perform adequately with any other shift in the underlying operating distribution. External or temporal validation will be needed to verify the ability of our model to generalize to new banks or changes in the client population.

Predictive accuracy aggregated in one number can mask heterogeneity of predictive performance across groups. Predictive accuracy of the algorithm by age bands, activity, gender, or geography needs to be considered in light of the prevalence and sample size in each group. There are various error-rate based criteria, such as equal opportunity and equalized odds which address different facets of predictive parity (Hardt et al., 2016), and calibration can contradict error-rate parity if group outcome prevalences are different (Pleiss et al., 2017). Direct input-sensitivity tests, where one recorded demographic feature is varied while other entries remain fixed, reveal dependence of the fitted score on that input but should not be interpreted as feasible causal counterfactuals.

Ultimately, computational aspects must become relevant at some point in any evaluation. Issues of training speed, inference delay, artifact size, complexity of the model, memory consumption, and explanation costs will indicate whether the gains in discrimination achieved are worth the trouble. The comparative consumption of resources must set the context of the evaluation, separating those that take place in the first locked evaluation from those in further frozen configuration evaluations. Otherwise, efficiency differences might be taken as performance features of the system. Ideally, one would want the model to make predictions with empirically plausible accuracy in view of the resources consumed.

2.6 Related Empirical Studies and Research Gap

2.6.1 Evidence from banking churn research

Previous work conducted on retail banking by Mutanen et al. (2010) has shown the existence of the customer churn prediction problem for individual customers. De Lima Lemos et al. (2022) used various methods to examine a set of data associated with financial organizations and have found Random Forest superior to other approaches. Both works show the relevance of predictive learning to bank-customer relations; however, their results depend on certain datasets, algorithms, and evaluation design, which makes them incomparable and should not be used as benchmarks for one another.

Some more recent works have extended the pipeline of model construction. Brito et al. (2024) considered data preparation, imbalance resolution, hyperparameters optimization, and XGBoost for a large retail banking dataset, showing that the result cannot depend solely on the classifier being used. In the works of Văduva et al. (2024), banking customer churn detection was addressed using Random Forest and LightGBM with probability calibration and SHAP; their work emphasizes the importance of the probability quality and interpretability, but does not provide a direct comparison between several robust gradient-boosting techniques and modern deep tabular architectures.

Results obtained in other service industry domains inform on methodological considerations but do not provide direct comparability with banking metrics. The telecom industry has explored the impact of alternative oversampling approaches, cross-companys transfer, and data transformations (Amin et al., 2016; 2019). These considerations demonstrate how the effectiveness of churn prediction depends on class handling and differences in populations. As methods differ between event definitions, measurement procedures, and feature spaces, they can be used for justifying experiments only.

2.6.2 Gap and position of the present study

From the literature, there are four interconnected methodological considerations. The first one is that the comparison of the banks should take into account well-working baseline gradient boosting in addition to the current deep models and not draw conclusions on the superiority of neural networks from poor baselines. The second one is that tuning, balancing, calibration, thresholding, and evaluation should be performed in a leakage-controlled pipeline in such a way that the model ranking would be interpretable. The third one is that uncertainty and paired statistical inference should be used for comparing competitive models in case when the difference in their scores is small.

The current study tackles these requirements with the help of an integrated benchmark that includes five classical learning algorithms and four deep tabular learners for the same customer-level banking task. The major contribution of the work is methodological evidence that relies on a controlled comparison, statistical analysis, cross-method explanations, and robustness analysis. This work does not provide a novel neural network architecture, causal analysis of factors responsible for the departure, proof of discrimination, or deployment readiness. The following chapter translates these

conceptual requirements into a specified dataset, training protocol, evaluation design, and reproducibility record.

CHAPTER 3. RESEARCH DATA AND METHODOLOGY

3.1 Research Design and Overall Framework

3.1.1 Research strategy and evaluation boundaries

This experiment uses the controlled comparative design for prediction of churn among bank customers. The objective of this study is to compare the performance of five traditional machine learning models and four deep tabular learners when using the same target, predictors, population development, and final customer-level holdout. Instead of suggesting an innovative classifier, the study defines a process of benchmarking and validation, where discrimination, probability quality, decision thresholds, uncertainty statistics, interpretability, robustness, sub-grouping characteristics, and computational complexity will be assessed as related but distinct model performance measures.

The key protective measure is the separation of model development and final evaluation. Immutable 20% holdout is generated prior to any tuning and calibration, while the rest 80% of cases will serve as the only source of information for all development-stage choices. Pre-processing is trained on appropriate training splits and not on data to be used to evaluate those transformations. This order mitigates the possible problems of leakage, which can artificially enhance classification performance (Kapoor & Narayanan, 2023). Moreover, it prevents the holdout outcomes from affecting the choice of hyperparameters, neural components, class imbalance adjustment, calibration and decision thresholds.

The method relies on a stratified 80/20 train-test split, hyperparameter tuning for each model separately, and standard repeated stratified cross-validation approach. All models are fitted once in the fixed train setup and tested on the same 2,000 examples. Probabilities and classifications are saved before any tests or post-hoc analysis of models is done. Thus, the tests, SHAP analysis, subgroups analysis, and resources assessment describe the fixed models behavior, and do not give any possibilities to fit them again.

3.1.2 Seven-phase benchmarking workflow

Figure 1 shows the seven stages of the research pipeline. Stage 1 checks data fitness and integrity, makes the development-holdout stratified split, and stores the fingerprints of the data. Stage 2 does model development iteratively solely on the 8,000 customer development slice, including tuning, validations, ablations, and specific probability calibration and thresholds. Stage 3 trains the frozen models and opens the holdout slice for the single prediction phase.

The rest of the stages analyze the fixed products and estimators. Stage 4 computes the uncertainty and paired tests on the predictions on holdout data that remain untouched. Stage 5 measures the attributions of features and consistency of explanation ranking between representative models. Stage 6 evaluates frozen models under noisy data, subgroup performance, and direct sensitivity to selected demographic features. Stage 7 tracks the computational performance and reproducibility evidence, including post-test

process memory measurements. The stages make up the one-way pipeline where the information from Stages 3 to 7 do not feed back to change any decision in Stage 2.

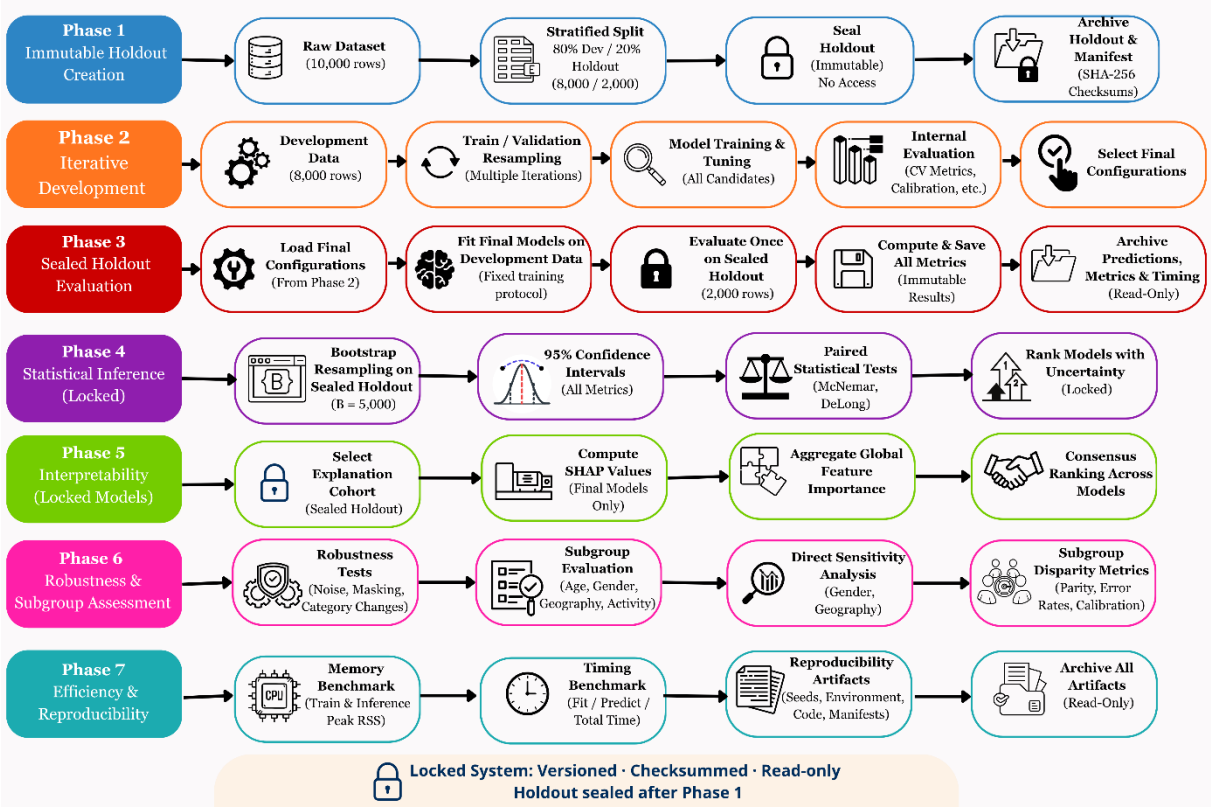


Figure 1. Leakage-Controlled Benchmarking and Validation Framework

3.2 Dataset and Feature Specification

3.2.1 Source, target definition, and feature eligibility

The analysis utilizes the publicly available Bank Customer Churn data set provided through Kaggle (Akturk, 2020). It consists of 10,000 samples of customers and 14 original columns. The binary target variable Exited equals one if the record shows that the customer has churned and 0 otherwise. There are 2,037 churned customers in the data, meaning that the class imbalance equals 20.37%, which necessitates accuracy-independent evaluation metrics.

Three of the original columns were not used to form predictors. The column RowNumber is the number of the row in the source file, while the CustomerId is an identification number, and Surname refers to an individual name; the inclusion of the latter does not make sense in this benchmark. The CustomerId column is used solely for matching the records to ensure reproducibility; it is not used by any predictive model.

None of the following predictors is included in the above list: complaint indicators, measures of customer satisfaction, reward points' history, bank card level, target-derived statistics, or customer transaction sequences. Label and features denote the data provided in the static table; no forward prediction period or independent observation of

a bank action is created in the dataset. This limitation matters once the model is considered for retrospective churn prediction or used as a prototype of research rather than a predictor of retention effectiveness.

3.2.2 Data audit and partition characteristics

The initial data audit evaluated schema consistency, duplicates, missing or infinite values, constant predictors, distributional balance of classes, and whether any of the identifiers marked as not predictors had leaked into predictors accidentally. The resulting research dataset included no missing or infinite values, no duplicates, and no constant predictor. No imputation of missing values was required for ordinary training of the models. Any imputation-like replacements are performed solely as part of subsequent robustness stress testing, where absence of some inputs is artificially created.

A stratified fixed 80/20 split using random seed 42 yielded 8,000 observations for development and 2,000 for lock-down test set. Class distribution for each of those sets is listed in Table 1. It is important to maintain roughly the same distribution of classes, as precision, PR-AUC, and operating point statistics partially depend on the churn rate. The hold-out set is randomly sampled from the same source as the training set without being from another source, region or future time frame.

Table 1. Dataset and Experimental Split Characteristics

Component	**Value**
Total observations	10,000
Predictive features	10
Non-predictive fields excluded	3
Churned customers	2,037
Overall churn prevalence	20.37%
Development observations	8,000
Development non-churners	6,370
Development churners	1,630
Development churn prevalence	20.38%
Locked-test observations	2,000
Locked-test non-churners	1,593
Locked-test churners	407
Locked-test churn prevalence	20.35%
Random seed for the initial split	42

Note. Churn corresponds to Exited = 1. The test partition remained sealed during all model-development activities.

3.3 Data Partitioning and Preprocessing

3.3.1 Fold-local transformations and categorical representation

In the case of a benchmark comparable to others with distinct numeric codes for each algorithm with a distinct input interface, it is essential that all algorithms should get ten conceptually equivalent predictors, each transformed in accordance with its algorithmic implementation. For estimators that need numeric design matrices as inputs, categorical predictors are appropriately converted into numeric form; CatBoost uses natural categorical representation. Continuous predictors are scaled for those estimators whose performance depends on the scale of predictors, such as Logistic Regression and MLP. Tree-based algorithms do not require any universal scaler.

The crucial requirement of comparability is that both encoder or scaler object must be fitted using only the part of data available to the specific fitting routine. In cross-validation, preprocessing of each algorithm is performed on training observations of the specific fold and applied to observations of validation fold. None of the preprocessing parameters are learned from the data of the out-of-fold validation partition. This requirement is fulfilled in the same way for the sealed test set.

This means that the preprocessing pipeline must be considered a part of the model and not as a globally-applied process that is done before splitting. The same rule applies when considering sampling for class rebalancing: An oversampler cannot use validation samples to generate synthetic training samples. All evaluation rows maintain their labels and frequencies.

3.3.2 Holdout locking and preserved evaluation artifacts

Creation of the holdout indices occurs just once. After that step, development routines can be applied repeatedly on the set of 8,000 qualified data points, but the holdout will no longer be a part of validation folds, tuning of hyperparameters, choice of an early stopping epoch or of probability calibration map. The same set of indices will be used later to synchronize predictions of all nine models on a customer-by-customer basis.

Once final training and calibration settings are fixed, the test split is scored once per model that was trained. Raw probability of churn, calibrated probability, locked binary predictions, chosen cutoffs, indices, metrics, and timing are stored in the archive of immutable results. It is critical for proper paired testing – predictions for each model are scored against the same people, and further analysis can use that result without repeating prediction function estimation.

3.4 Benchmark Models and Hyperparameter Optimization

3.4.1 Nine comparison models

There are nine classifiers that represent different assumptions and architectures. Classical models are Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost. The former provides an additive linear benchmark; the latter is a bagged non-linear decision tree. Three boosting approaches build sequential ensembles of trees that

differ in their computational and categorical data treatment approaches (Breiman, 2001; Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018). The choice of these three makes it impossible to evaluate modern deep learners by comparing them against a poor linear model only.

Four deep tabular models are a tuned MLP, TabNet, FT-Transformer, and RealTabR. The former corresponds to a traditional feed-forward neural network; the second one relies on sequential feature selection mechanism; the third model embeds tabular features as feature tokens and learns feature interactions using self-attention mechanism; and the last one represents tabular learning approach based on the TabR framework (Arik & Pfister, 2021; Gorishniy et al., 2021, 2024). All nine models output continuous predictions of the probability of Exited=1 before the application of their respective locked decision rules.

Benchmark does not imply the implementation of any novel architecture or any assumption about the similarity of optimization methods used internally by algorithms. It consists of the application of a proven optimization method specific to the model and a standard comparison procedure at the development stage, as well as one shared locked test set.

3.4.2 Hyperparameter search strategy and classical-model configurations

The hyperparameter optimization process takes place with Optuna utilizing Tree-structured Parzen Estimator sampler seeded with 42 (Akiba et al., 2019). The goal is to maximize ROC-AUC for candidate selection. Classical algorithms get 30 trials of optimization using fixed stratified five-fold cross-validation, whereas neural networks have 15 trials each of them with fixed stratified three-fold cross-validation. The restricted search space budget for deep learning algorithms is an explicitly stated computational limitation but not proof of equal tuning effort for different types of algorithms.

Table 2 contains the exact search space and values of selected hyperparameters for the five classical algorithms. Parameters which are not highlighted as tuned use default values according to the implementation documentation. Logistic Regression employs the liblinear solver and maximum 5,000 iterations. Random Forest, XGBoost, LightGBM and CatBoost fit the model with random seed 42 and use four CPU cores if available. XGBoost uses log loss as training metric and CatBoost uses log loss as objective and AUC as evaluation metric. The configuration table is left in the body as these parameters are required for understanding the benchmark itself and not additional results.

Table 2. Classical Model Search Spaces and Selected Configurations

Model	Hyperparameter	Search space	Selected value
Logistic Regression	(C)	10^{-4} to 10^{3}, log scale	0.063915726
Logistic Regression	Penalty	{L1, L2}	L1

Model	Hyperparameter	Search space	Selected value
Random Forest	Maximum depth	{None, 5, 8, 12, 16, 20}	20
Random Forest	Number of trees	200–600, step 50	450
Random Forest	Minimum samples to split	Integer 2–20	16
Random Forest	Minimum samples per leaf	Integer 1–12	12
Random Forest	Maximum features	{sqrt, log2, 0.5, 0.8}	0.5
XGBoost	Number of trees	150–700, step 50	250
XGBoost	Maximum depth	Integer 2–8	5
XGBoost	Learning rate	0.01–0.25, log scale	0.045438329
XGBoost	Minimum child weight	1–15, log scale	3.243329533
XGBoost	Subsample ratio	0.6–1.0	0.732972223
XGBoost	Column subsample ratio	0.6–1.0	0.930320661
XGBoost	Gamma	0–5	4.001822689
XGBoost	L1 regularization	10^{-8}–5, log scale	2.1997×10^{-7}
XGBoost	L2 regularization	10^{-3}–20, log scale	0.001056813
LightGBM	Maximum depth	Integer 3–10	6
LightGBM	Number of trees	150–700, step 50	500
LightGBM	Number of leaves	7 to min(127, $2^{\text{depth}} - 1$)	35
LightGBM	Learning rate	0.01–0.25, log scale	0.010402388
LightGBM	Minimum child samples	Integer 10–100	43
LightGBM	Subsample ratio	0.6–1.0	0.828958691
LightGBM	Column subsample ratio	0.6–1.0	0.668709834
LightGBM	L1 regularization	10^{-8}–5, log scale	1.5702×10^{-7}
LightGBM	L2 regularization	10^{-3}–20, log scale	0.005553505
CatBoost	Iterations	200–700, step 50	650
CatBoost	Depth	Integer 4–10	4
CatBoost	Learning rate	0.01–0.25, log scale	0.017590205
CatBoost	L2 leaf regularization	1–20, log scale	8.147624937
CatBoost	Random strength	10^{-3}–5, log scale	0.002499942
CatBoost	Bagging temperature	0–5	2.061263810

Model	Hyperparameter	Search space	Selected value
CatBoost	Border count	{32, 64, 128}	64

Note. Search spaces and selected values originate from the frozen Optuna configuration records. Fixed controls are described in Section 3.4.2.

3.4.3 Deep tabular architectures and final training controls

The MLP tune includes the hidden layers' architecture, activation, dropout, Batch Normalization (optional), learning rate, weight decay, and batch size. The chosen hidden widths are 256-128-64 with SiLU activation, dropout 0.242270567, no Batch Normalization, and AdamW as the optimization algorithm. The used loss is binary cross-entropy with logits. The locked fit went for 35 epochs due to development-only early stopping. Those are the actual parameters of the tuned network, not the ideal generic MLP.

TabNet configuration consists of decision and attention widths, attentive decision steps, sparsity penalty, categorical embedding size, gated linear-unit layers, learning rate, and batching options. FT-Transformer is defined via token width, number of attention heads, depth of transformer blocks, feedforward expansion ratio, independent dropouts, optimizer options, and scheduling options. RealTabR configuration includes representation width, number of encoder and predictor blocks, retrieval context size, dropouts, and fixed internal optimizer options. Table 3 contains all those tunable parameters' ranges and selected values, fixed training options, and training results from those four models.

Neither the number of epochs nor the criteria for stopping should be seen as an indication of a fixed training budget that is applied uniformly. Neural networks employ early stopping based on development data; the patience and maximum epochs vary according to model architecture. While the final models use the same development data and the same early stopping partition for deep learning, there are different times spent on fitting and trainable parameters.

Table 3. Deep Tabular Model Search Spaces and Final Configurations

Model	Hyperparameter	Search space or control	Selected value
MLP	Hidden structure	{64-32, 128-64, 128-64-32, 256-128-64}	256-128-64
MLP	Activation	{ReLU, GELU, SiLU}	SiLU
MLP	Dropout	0–0.50	0.242270567
MLP	Batch norm.	{True, False}	False
MLP	Learning rate	10^{-4} to 3×10^{-3}, log scale	6.73704×10^{-4}
MLP	Weight decay	10^{-6} to 10^{-2}, log scale	3.68226×10^{-5}
MLP	Batch size	{128, 256, 512}	256
MLP	Optimizer	Fixed	AdamW

Model	Hyperparameter	Search space or control	Selected value
MLP	Loss function	Fixed	BCEWithLogitsLoss
MLP	Max epochs	Fixed	80
MLP	Early-stop metric	Fixed	ROC-AUC
MLP	Early-stop patience	Fixed	10
MLP	Min. improvement	Fixed	10^{-5}
MLP	Completed epochs	Early-stopping outcome	35
MLP	Trainable params.	Recorded	44,801
TabNet	Decision width (n_d)	{8, 16, 24, 32}	16
TabNet	Attention width (n_a)	Set equal to n_d	16
TabNet	Decision steps	Integer 3–6	3
TabNet	Gamma	1.0–2.0	1.815461428
TabNet	Sparse reg.	10^{-6} to 10^{-2}, log scale	6.72093×10^{-4}
TabNet	Cat. embed. dim.	{1, 2, 3, 4}	2
TabNet	Independent GLU layers	Integer 1–3	1
TabNet	Shared GLU layers	Integer 1–3	3
TabNet	Learning rate	0.001–0.03, log scale	0.008330804
TabNet	Batch size	{128, 256, 512}	128
TabNet	Virtual batch size	Batch size / 4	32
TabNet	Optimizer	Fixed	Adam
TabNet	Mask type	Fixed	sparsemax
TabNet	Max epochs	Fixed	100
TabNet	Early-stop metric	Fixed	ROC-AUC
TabNet	Early-stop patience	Fixed	12
TabNet	Completed epochs	Early-stopping outcome	57
TabNet	Trainable params.	Recorded	15,738
FT-Transformer	Token dimension	{32, 64, 128}	128
FT-Transformer	Attention heads	{4, 8}	8
FT-Transformer	FFN expansion factor	1.0–2.5	2.043330601
FT-Transformer	Transformer layers	Integer 1–4	2

Model	Hyperparameter	Search space or control	Selected value
FT-Transformer	Attention dropout	0–0.3	0.171300229
FT-Transformer	FFN dropout	0–0.3	0.088072457
FT-Transformer	Residual dropout	0–0.2	0.041619222
FT-Transformer	Learning rate	10^{-4} to 3×10^{-3}, log scale	2.41935×10^{-4}
FT-Transformer	Weight decay	10^{-6} to 10^{-3}, log scale	1.80805×10^{-5}
FT-Transformer	Batch size	{128, 256, 512}	128
FT-Transformer	Optimizer	Fixed	AdamW
FT-Transformer	Max epochs	Fixed	60
FT-Transformer	Early-stop patience	Fixed	8
FT-Transformer	LR scheduler	Fixed	Enabled
FT-Transformer	LR scheduler patience	Fixed	4
FT-Transformer	Final training seed	Fixed	2027
RealTabR	Main rep. width	{64, 128, 256}	64
RealTabR	Width multiplier	{2, 4}	4
RealTabR	Encoder blocks	Integer 1–3	1
RealTabR	Predictor blocks	Integer 1–3	2
RealTabR	Context size	{16, 32, 64}	64
RealTabR	Context dropout	0–0.3	0.085074445
RealTabR	Dropout 0	0–0.3	0.124326502
RealTabR	Dropout 1	0–0.3	0.155657282
RealTabR	Batch size	{128, 256, 512}	512
RealTabR	Optimizer	Fixed internal	AdamW
RealTabR	Optimizer LR	Fixed internal	3.12127×10^{-4}
RealTabR	Optimizer WD	Fixed internal	1.22604×10^{-6}
RealTabR	Max epochs	Fixed	40
RealTabR	Early-stop patience	Fixed	6
RealTabR	Eval. batch size	Fixed	512
RealTabR	Candidate-enc. batch size	Fixed	512
RealTabR	Final training seed	Fixed	2027

Note. The entries include tunable hyperparameters, fixed controls, and recorded training outcomes. Values are not recomputed.

3.5 Development-Stage Validation and Controlled Ablations

3.5.1 Repeated cross-validation after model-specific tuning

After model-specific selection with Optuna, each tuned algorithm is subjected to repeated stratified five-fold cross-validation with three repeats on the development dataset. This results in 15 validation scores for each model using the same evaluation methodology, independent of the folds that were used during the hyperparameter optimization process. Keeping the latter evaluation methodology fixed lowers reliance on a random validation split, and allows for comparing the different tuned models on the same development set.

The ranking measure of choice is ROC-AUC. PR-AUC, Brier score, expected calibration error (ECE) and fitting time offer supplementary information regarding minority event detection, probability accuracy and computational cost. The repeat cross-validation constitutes the development-level model comparison, and must not be mistaken for a cross-validated ensemble of the final holdout test set: the test probabilities will come from a single fit of each model.

3.5.2 Class-imbalance ablation

A development-only experiment with three class-imbalance approaches is done to compare: keeping the class frequency the way it is, applying the class weighting and SMOTENC synthetic oversampling. SMOTENC is used for numerical and categorical data combined. Synthetic data are produced only in a training set, whereas validation sets keep their true class frequencies (Chawla et al., 2002). This separation is necessary for avoiding synthetic training observations being similar to the ones against which the classifier is evaluated.

In the experiment, both ranking metrics without thresholds and quality metrics with thresholds are considered. The increase in a positive class weight or the number of minority examples can lead to the recall increase at a default threshold but also change the precision and posterior scores calibration. Hence, the decision on the approach to apply is made based on the development data as a whole rather than one recall metric only. These results and the final decision are presented in Section 4.3.

3.5.3 MLP component ablation

The factorial MLP experiment studies dropout and Batch Normalization by analyzing four different setups: without both, dropout only, Batch Normalization only, and with both. This is done prior to releasing the holdout to see whether standard neural network layers help with generalization given the training setup documented above. The chosen MLP model and fixed variables are summarized in Table 3; the rationale for choosing this architecture belongs to Chapter 4.

SHAP is not regarded as a layer, a loss term, a feature, or an extra predictive model in this ablation. SHAP is used only as an interpretation technique after fitting the model.

Hence, no improvement in the prediction performance could be linked to the usage of SHAP as part of the architecture.

3.6 Probability Calibration, Threshold Selection, and Final Training

3.6.1 Model-specific probability calibration

The classifier will rank the clients according to the level of risks without necessarily producing probabilities which systematically overestimate or underestimate the observed event rates. The pipeline clearly differentiates between the ranking capability and the quality of probabilities produced. Based on the out-of-fold predictions made on the development dataset, each model undergoes assessment based on the three calibration options, which are no transformation, sigmoid calibration, and isotonic calibration (Niculescu-Mizil & Caruana, 2005; Guo et al., 2017).

Calibration is done based on the set criteria where the focus is on Brier score and log loss on development dataset. The transformation used is associated with the particular model used, and it is locked before evaluation on the test dataset. The process gives room for choosing different calibrations since the probability distortion of the models may not be the same. An improvement on the development dataset does not mean that the same applies on any finite holdout sample. An unfavorable calibration result on the holdout dataset does not give room for changing the calibrator used.

3.6.2 MCC-based operating threshold

Once the calibration problem is solved, the classification threshold is chosen based on maximizing the Matthews correlation coefficient (MCC) on the predictions made during out-of-fold development. The MCC is calculated using true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) and takes into account all four cells of the confusion matrix (Chicco & Jurman, 2020). Thus, threshold selection is independent of both hyperparameter tuning and the final hold-out calculation.

Each of the models gets an individual threshold, rather than a common 0.50 cut-off point. Such a design allows different calibration probabilities and score distributions to yield different locked binary predictions. While ranking criteria can be evaluated independently from the threshold used, the reported recall, F1, specificity, balanced accuracy, and MCC depend on the thresholds that were chosen beforehand. In the application prototype, the XGBoost threshold of 0.392 is reused without re-tuning; its meaning is described in Chapter 5.

3.6.3 Final model fitting and immutable outputs

All five classical estimators are fit to all 8,000 development samples using the same frozen hyperparameter and preprocessing settings. The final deep learning fit uses a stratified split from the development set into 6,800 gradient training samples and 1,200 samples for early stopping. This latter split is held constant using random seed 2027. The early stopping samples are not included in the sealed test set, and the differential in the sample size of gradient training observations is maintained for the computational comparison.

This results in the generation of the final fitted instance of each algorithm, which provides both raw and, where appropriate, calibrated probabilities for the 2,000-holdout customer sample. The locked thresholds are then applied to convert these probabilities to a binary classification. There is no recalibration or threshold selection on the holdout, nor is the holdout used to select an alternate model based on its prediction errors.

3.7 Evaluation Metrics and Statistical Inference

3.7.1 Discrimination and probability-quality measures

ROC-AUC serves as the discrimination statistic since it captures, independently of any chosen thresholds, the degree to which predicted values for the churners are larger than those for the non-churners. Precision-recall AUC is used, together with ROC-AUC, as, when about one out of five customers is in the churn class, there needs to be a differentiation between positive-class recall and ranking discrimination (Saito & Rehmsmeier, 2015). No single measure proves that the resulting numbers predict the probability of the event accurately.

Accuracy of the predicted probability values is assessed using the Brier score, defined as the average square difference between predicted and observed probabilities (Brier, 1950). The log loss places a higher weight on the inaccurate and confident predictions. Ten-bin ECE describes the differences between average predicted probabilities and observed frequencies of churn in probability bins. ECE is interpreted alongside Brier score and log loss due to the descriptive approximation implied by binning.

For n customers with observed labels y_i and predicted probabilities p_i, the Brier score is $n^{-1}\Sigma_i(p_i - y_i)^2$. For probability bins $B_1,\ldots,B_{10}$, ECE is $\Sigma_m(|B_m|/n)\cdot|acc(B_m) - conf(B_m)|$, where acc denotes the observed event frequency and conf the mean predicted probability in that bin. A lower Brier score, log loss, and ECE show better probability calibration based on their own definitions.

3.7.2 Locked-threshold classification and capacity-based ranking

The performance report provides accuracy, precision, recall, specificity, F1-score, balanced accuracy, and MCC for the locked binary decision set. The aforementioned metrics take into account different costs and balancing between false positives and false negatives: recall is about the percentage of real churners identified, specificity is about the percentage of actual non-churners rejected, and balanced accuracy takes the mean of sensitivity and specificity. MCC is kept as a measure of all confusion matrix values and not as an alternative name for F1-score.

The separate top-20% ranking evaluation demonstrates the scenario where the operator has to consider only one-fifth of the customers. Upon ranking customers by their predicted probabilities, the 20% recall will be the ratio of all churners to those that appear among the top-ranked fifth. Lift at 20% evaluates the ratio to the baseline prevalence. The aforementioned ranking criterion is not the same as the locked MCC threshold. Top capacity ranking may include customers below the binary decision threshold, and threshold-based flagging is not limited to the capacity.

3.7.3 Confidence intervals, paired tests, and multiplicity

Uncertainty is based on 5,000 bootstrap samples of the fixed holdout cases, where the same resampled customer indexes are used for competing models (Efron, 1979). Such paired sampling takes into account the correlation that exists between two model scores resulting from the same actual observations, rather than separate evaluation populations. The 95% confidence intervals reported here capture sampling variability, conditional on the data and experimental setup, but do not include the institutional changes, time effect, or measurement shift not present in the benchmark.

Differences between ROC-AUC values are tested by DeLong test for correlated receiver-operating-characteristic areas (DeLong et al., 1988). The paired bootstrap comparison accounts for differences in PR-AUC, Brier score, log loss, F1-score, balanced accuracy, MCC, and recall@20%. Exact McNemar test is applied to differences in the locked binary classifications. There are 36 different model pairs with 9 models. P-values in each metric group are adjusted using Holm's procedure (Holm, 1979).

Small unadjusted p-value and a numerical ranking are not exchangeable for a solid general superiority. Conclusions from statistics are made solely after predefined pairing and multiplicity adjustment; statistical nonsignificance is understood as an absence of sufficient data about a distinction, but not as evidence that the two models are identical. No statistical procedures provide any new information about the frozen model definitions.

3.8 Explainability Validation Protocol

3.8.1 Feature redundancy assessment

Prior to comparing attribution for features, predictor redundancy is measured, which can reassign explanation scores among predictors exhibiting some form of correlation. The audit makes use of Pearson and Spearman correlation for numeric relationship analysis, variance inflation factors as a numerical design matrix metric, design condition number for a standardized design matrix, and adjusted Cramér’s V for categorical relationships. These various tests account for various types of relationships and facilitate interpretation of the SHAP rankings without assuming that low linear correlation precludes non-linear relationships (Aas et al., 2021).

Redundancy testing is an interpretive tool and not a feature selection method performed after hold-out release. All ten valid predictors are retained in the compared predictive input space. The audit results are shown prior to the SHAP results in Chapter 4.

3.8.2 Architecture-independent SHAP and cross-model agreement

The main focus of explainability is the four locked models, namely, XGBoost, CatBoost, MLP, and FT-Transformer. The standard explanation pipeline is the permutation-SHAP approach to explain the end calibrated churn probability instead of a mixture of feature importances, which may have different measurement units or native definitions of importance in each architecture (Lundberg & Lee, 2017). The explanation is provided

for the stratified group of 300 holdout instances using a stratified training dataset of 100 observations. The utilized explanation budget is five permutations with 105 model evaluations per each explained instance.

For each model, the global importance is calculated as the average absolute SHAP value among the common explanation group. Then, ten of those importance values are normalized for a particular model prior to the consensus across models. The importance ranking correlations are evaluated using Spearman and Kendall coefficients and the top three and top five consistent important features are found. Another 2,000 bootstrapped resamples of the explanation group are used to determine the stability of the rankings and frequency of the selections.

The directions of the features are analyzed independently of their global importances. Associations between the values of the observed features and signed SHAP contributions are studied based on quintiles of the continuous features and recorded values of the discrete or binary ones. Positive SHAP contribution indicates the increase of the model score with respect to its reference prediction for the chosen explanation; it does not imply any manipulation of the feature will change the customer behavior.

3.8.3 Supplementary TabNet and RealTabR fidelity checks

Explainability is also extended to the final TabNet estimator with the same cohort, background, ten-feature space, and permutation budget. Before attributing, the serialization of the pipeline is restored and is compared with the archived raw predictions. The largest absolute difference is measured to be 8.79×10^{-7}, while the additive SHAP residual is measured to be 1.48×10^{-7}, meeting the set fidelity threshold of 10^{-5}. This is evidence that attribution was done to the original final TabNet estimator and not a significantly different re-fit.

A greater caution is required when explaining the RealTabR estimator. The original fitted estimator is not serializable due to an internal local lambda function in the code. The reconstructed estimator with frozen parameters, the same exact 6,800/1,200 training partition, and seed value of 2027 fails to recover the original fitted estimator's final raw probabilities within the set tolerance of 10^{-5} with a maximum absolute difference of 0.2373. Therefore, SHAP values calculated from the reconstructed estimator do not pertain to the final fitted RealTabR estimator.

3.9 Robustness, Subgroup, and Computational Evaluation

3.9.1 Input-perturbation robustness

Locked versions of the XGBoost, CatBoost, MLP, and FT-Transformer models are tested under seven input stress conditions: mild and moderate noise; 5% and 10% simulated missingness using median/mode imputation; 5% and 10% categorical stress; and a mixed stress condition. Numerical stress is applied to CreditScore, Age, Tenure, Balance, and EstimatedSalary. Integer variables are rounded and clamped within the bounds seen during development. Categorical stress substitutes Gender and/or

Geography with a different valid category, while masking uses the median or mode learned during development.

Each test scenario is run 20 times with consistent seeds, but the same fitted estimator, calibrated model, and threshold parameters are used throughout. Evaluation includes changes in ROC-AUC, PR-AUC, Brier score, F1 score, balanced accuracy, and MCC scores, as well as probability change, rank correlation change, and the proportion of classifications that change due to locking. This test assesses the sensitivity to these particular forms of imperfect input. It does not simulate a natural distribution shift or demonstrate robustness to any other form of degradation.

3.9.2 Subgroup performance and direct demographic sensitivity

Sub-group analysis employs immutable hold-out outputs from all nine models. Sub-groups are based on Gender, Geography, activity status, and four age ranges (18-34, 35-44, 45-54, and 55+). Each sub-group is provided with the churn observed prevalence, average prediction probability, predicted-positive rate, ROC-AUC, PR-AUC, Brier score, calibration-in-the-large, precision, recall, false positive rate, specificity, F1, balanced accuracy, and MCC. Disparity statistics include demographic parity, equal opportunity, false positive rate, equalized odds, and predictive parity gaps, along with calibration and Brier intervals. The uncertainties of such disparity statistics are estimated with 1,000 bootstrap iterations (Hardt et al., 2016; Pleiss et al., 2017).

Another related direct-sensitivity analysis focuses on Gender and Geography for the four illustrative models. For each customer, one of the audited categorical variable values is changed while keeping all others constant. Gender comparisons involve comparing Female with Male, while geography comparisons involve assigning France, Germany, and Spain to each customer, with measurements of record-wise probability differences, probability intervals, and locked decision changes. Mean sensitivities receive 2,000-bootstrap confidence intervals. These interventions diagnose direct predictor dependence; they are not causal counterfactuals, conclusive legal fairness tests, or evidence about the effect of changing an individual's real-world identity or location.

3.9.3 Computational cost and post-test memory measurement

The efficiency measurements document the time to fit each final estimator, time to predict 2,000 holdout instances, throughput, latency per instance, artifacts size kept, complexity in terms of trees/parameters available and permutation-SHAP explainability time. All timings from the original fit and inferencing come from the timing in the four-core CPU-only setup. The computational comparison assumes these implementation details and resources, particularly latency comparisons for extremely short tasks.

Peak memory usage is separately replicated in a different post-test exercise with the model being frozen at its configuration. Each model is run as a standalone process, where process tree resident set size (RSS) is sampled every roughly two milliseconds during fitting and holdout inferencing. Absolute peak RSS usage as well as peak increment relative to the baseline process usage are retained. Replications use the 8,000

instance classical fitting protocol or 6,800/1,200 deep fitting protocol depending on the model.

Since process-memory measurement came after locked predictive evaluation, the results measure the memory consumption of the frozen model instance and not the specific model instance in question. The estimated models do not consistently generate immutable probabilities, and thus predictions based on them are never used to substitute for the benchmark score. The memory results are analyzed in this context in Chapter 4 and not as a new independent form of predictive evaluation.

3.10 Experimental Environment and Reproducibility

All locked experiments take place within a Kaggle Notebook environment on a Linux x86_64 machine with Python 3.12.13, four logical CPU cores, and approximately 31.35 GB of RAM. No GPU or CUDA capabilities are enabled. Table 4 summarizes key software versions, including numerics, models, Optuna, SHAP, and deep-learning libraries. The original CPU identifier is unknown and not estimated. Claims of reproducibility therefore rest upon the actual environment information rather than any hypothetical hardware configuration.

The original split, optimization sampler, model parameters, fitted preprocessing objects, calibrators, thresholds, random seeds, and predictions are stored via configuration files and experiment outputs. Hash signatures uniquely identify the source dataset, train/holdout split, uncalibrated and calibrated probability matrices, and final performance metrics. The initial split and Optuna sampler are seeded with 42; deep-learning model training is seeded with 2027 for the 6,800/1,200 fixed train/test split and model-dependent processing. Such data ensure traceability of the reported pipeline despite the inherently stochastic nature of machine learning and specific implementation differences at the library level.

The guiding rule is that post-hoc computational or inferential analyses will not affect the confirmatory yardstick. Chapter 4 uses the methodology to present empirical results, while Chapter 5 focuses on how a particular frozen model might be used to develop an illustrative churn risk decision-making interface.

Table 4. Experimental Environment and Software Configuration

Component	Configuration
Execution environment	Kaggle Notebook
Platform	Linux 6.12.90+, x86_64, glibc 2.35
Processor	Kaggle CPU runtime, 4 logical CPU cores
System memory	31.35 GB
GPU	None
CUDA	Not available

Component	Configuration
Python	3.12.13
NumPy	2.0.2
pandas	2.3.3
SciPy	1.16.3
scikit-learn	1.6.1
PyTorch	2.10.0+cpu
XGBoost	3.2.0
LightGBM	4.6.0
CatBoost	1.2.10
Optuna	4.9.0
SHAP	0.51.0
imbalanced-learn	0.14.1
pytorch-tabnet	4.1.0
PyTabKit	1.7.3
FAISS-CPU	1.14.3
Initial split seed	42
Optuna sampler seed	42
Hardware acceleration	CPU-only

Note. Software versions and resources reflect the locked study environment, not necessarily the separate application runtime.

CHAPTER 4. EXPERIMENTAL RESULTS AND ANALYSIS

This chapter presents empirical findings from the predefined development and holdout evaluations. The presentation will be done following the dependencies that were introduced in Chapter 3: input data verification is followed by development benchmark; development-only ablation and calibration choices are followed by holdout; and finally, the produced predictions allow for statistical inference, explainability, robustness, subgroup, and computational analysis. So, the result that was produced after the sealing of the holdout has been opened is treated as the property of the frozen systems rather than the chance to adjust their parameters or thresholds.

Three types of evidence are distinguished in this chapter. Results of repeated cross-validation represent the behavior of models during development. The fixed holdout with 2,000 customers represents the ultimate comparison of the predictive performance. Post-hoc analyses such as SHAP, synthetic perturbation, attribute sensitivity, and memory analysis at the configuration level represent specific properties of the frozen models.

4.1 Data Verification and Feature Redundancy

The input audit yielded 10,000 complete observations and the ten valid predictors listed in Section 3.2. There were no duplicate cases, missing or infinite values, or constant predictors. There were 2,037 occurrences of Exited as a positive attribute, i.e., 20.37% of the total dataset size. The relative proportions of the outcome were kept equal in both parts of the research: 1,630 churners out of 8,000 development records (20.38%) and 407 churners out of 2,000 holdout records (20.35%). Prevalence of the outcome will affect all of the subsequent analysis, as metrics like precision and PR-AUC depend on it.

This basic dataset audit does not prove that future operational records will be as complete. Instead, it provides limited methodological evidence that there was no existing baseline problem of missing data that needed a trained imputer. In turn, this allowed missingness introduction in the course of later research masking experiments. Similarly, elimination of RowNumber, CustomerId, and Surname predictors prevents any problems related to using identifiers of records.

Prior to comparing SHAP explanations, the study investigated if high dependence between inputs could result in attribution being distributed to redundant variables. The maximal absolute Pearson correlation coefficient for any pair of numeric predictors is 0.308, while the maximal absolute Spearman correlation coefficient is 0.320. Neither measure surpasses the 0.70 cutoff level. Small correlations between pairs imply there is little probability that the most significant numeric variables are merely duplicates of each other in linear or monotonic terms.

Multivariate and categorical diagnostics produced a coherent message. The maximal values of variance inflation factors are 1.341 for Geography_Germany and 1.337 for Balance, while none of VIF measures surpass five. The standardized design condition number is 1.867, while corrected Cramér's V is 0.021. This latter statistic implies a weak

correlation between the two measured categories. Diagnostic procedures did not detect multicollinearity or redundancy in common inputs representation.

It sets an appropriate background for the analysis of explanations, but does not validate the independence of SHAP values from dependencies. The importance of global features may still depend on the background, interactions, and perturbations outside the observed joint distribution (Aas et al., 2021). That is why the further discoveries of explanations are assessed for consistency in different models and resampling of explanations cohorts, not through redundancy analysis only.

4.2 Development-Stage Model Performance

4.2.1 Repeated cross-validation comparison

Following optimization of model-specific hyperparameters using Optuna, each resulting tuned estimator was tested using the same repeat-only, five-fold, stratified cross-validation approach with three repeats. Table 5 provides the average of the 15 results of the cross-validation process for each model. ROC-AUC and PR-AUC provide discrimination and positive-class retrieval metrics, respectively, while the Brier score and ECE assess the probability quality metrics with different definition.

The three gradient-boosted models have the best ROC-AUC performance. With the first-place score of 0.8672, XGBoost was succeeded by the models of CatBoost and LightGBM with scores of 0.8670. The difference between the scores of these models is no more than approximately two ten-thousandth, which means that their discrimination ability was indistinguishable at the shown precision. Similar is true for their PR-AUC performance, which has scores of 0.7055, 0.7046, and 0.7044, respectively. While XGBoost has the lowest Brier score among these three at 0.1015, LightGBM has the lowest ECE at 0.0184.

Table 5. Repeated Cross-Validation Performance on the Development Set

Model	ROC-AUC	PR-AUC	Brier score	ECE
XGBoost	0.8672	0.7055	0.1015	0.0190
CatBoost	0.8670	0.7046	0.1018	0.0224
LightGBM	0.8670	0.7044	0.1016	0.0184
Random Forest	0.8630	0.6921	0.1034	0.0211
FT-Transformer	0.8614	0.6910	0.1048	0.0294
MLP	0.8584	0.6935	0.1039	0.0185
RealTabR	0.8581	0.6883	0.1059	0.0396
TabNet	0.8427	0.6604	0.1097	0.0209
Logistic Regression	0.7634	0.4673	0.1366	Not retained as competitive

Note. Values are means over 15 development validation folds (five folds × three repeats); the holdout set was not used. ECE = expected calibration error. The Logistic Regression ECE is not retained in the source result summary.

FT-Transformer was the best deep tabular learner by ROC-AUC (0.8614), as opposed to MLP (0.8584), RealTabR (0.8581), and TabNet (0.8427). Nevertheless, MLP outperformed FT-Transformer by PR-AUC (0.6935 against 0.6910) as an example of how the order of models changes when the focus is shifted from discrimination to positive class retrieval. As for Brier score, MLP also did better than FT-Transformer (0.1039 versus 0.1048). In spite of all these differences, it remains true that the well-tuned boosting algorithms were superior development baselines in general.

Random Forest is another competitive algorithm occupying an intermediate place with mean ROC-AUC equal to 0.8630 and Brier score equal to 0.1034, while TabNet is inferior to other algorithms both in ROC-AUC and Brier score. Logistic Regression is the worst model among all considered ones in terms of ROC-AUC (0.7634), PR-AUC (0.4673) and Brier score (0.1366). The difference between the linear algorithm and nonlinear ones clearly shows that there is no enough evidence of linear additive dependencies between input features and target in the dataset.

The ROC-AUC and PR-AUC rankings have been illustrated in Figure 2. From the two figures, one can observe that the boost had a small edge over the best deep architectures in terms of development, but there was a considerable difference between the nonlinear algorithms and Logistic Regression. The graphed values are repeated validation means and cannot be interpreted as confidence intervals of the locked test results.

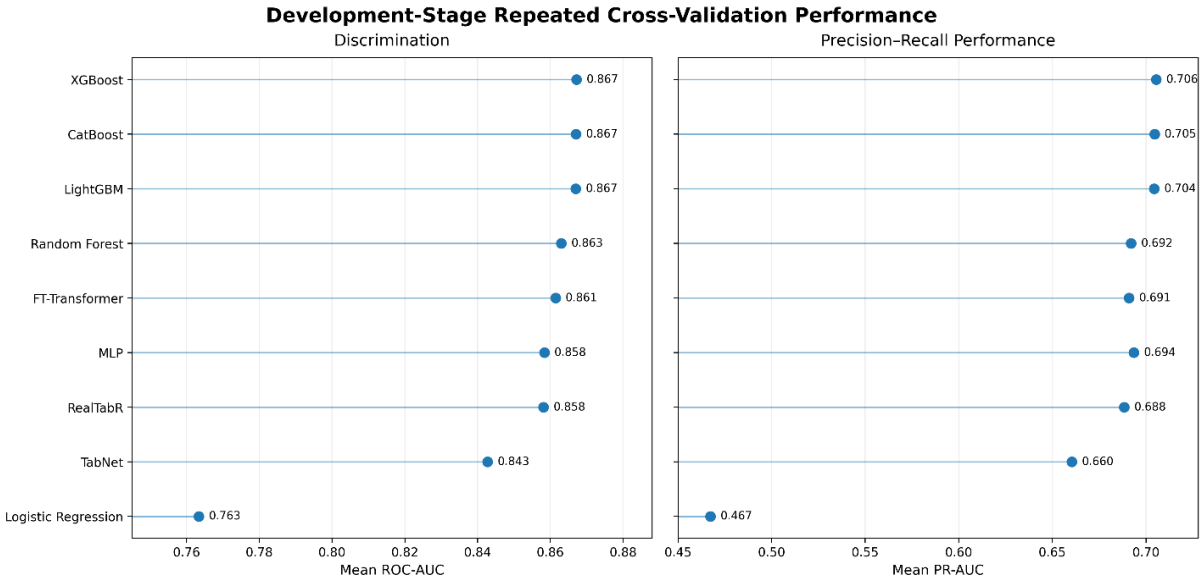


Figure 2. Development-Stage Cross-Validation Performance

4.2.2 What the development ranking can establish

The development results have addressed a specific question: which optimal models were providing competitive discrimination and sufficient probability quality according to the

same evaluation schedule? The results were not the ultimate benchmark tests of model performance. Model hyperparameters were tuned in advance using model-specific procedures, and the repeated folds were composed solely of 8,000 development examples. Therefore, the means calculated on repeated validation are useful to characterize model family stability in terms of available data but cannot replace the subsequent holdout sealed off.

It is also important to keep in mind that there is an asymmetry of training budget between classical and deep models. The former had 30 optimization trials evaluated on five folds and the latter 15 trials on three folds. This asymmetry has been pre-specified in the protocol of this research and is detailed in Chapter 3. This restricts a statement that the experiment has explored all possible architectures or provided identical optimization opportunities. Within the reported budget, however, the leading deep models did not demonstrate a persistent discrimination advantage over the boosted trees, consistent with the motivation for including strong tabular baselines (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022).

This means that the comparison will serve as the required starting point for the locked evaluation without assuming the outcome beforehand. The following section will discuss two modeling decisions made during the development stage, which might have affected the behavior of the holdout set: class imbalance and MLP structure. Probability calibration is discussed independently, since a good rank does not always provide good probabilities.

4.3 Ablation and Probability Calibration Results

4.3.1 Class-imbalance ablation

Positive-class prevalence was roughly one fifth of the data set. While the data set is imbalanced, it is not enough by itself to determine that either synthetic oversampling or class weight will enhance all assessment criteria. In order to find out how well each technique works, there was an experiment comparing training, class weight, and SMOTENC with resampling applied only on training folds. Validation folds remained as originally distributed.

With the default threshold of 0.50, both weighting and SMOTENC enhanced recall of churned clients. This enhancement happened with the lower quality of probabilities and less balanced confusion matrix in the experiment. It is interesting that a model can classify more customers as being at risk without changing their ranking and producing any valid probability estimates. In case of limited resources, it means only that the burden of the false positives increases.

The best performance development evidence, across ROC-AUC, PR-AUC, Brier score, ECE, and MCC was provided for both representative ablation learners, XGBoost and MLP, without any explicit weighting and/or synthetic oversampling. The final experimental protocol hence included neither of the two. It was simply an empirical choice in the context of the current distribution, predictors, and models. We do not make

any claims about how the class imbalance correction techniques (Chawla et al., 2002), like SMOTENC or class-weighting, may be generally negative.

It is important to recognize another important implication of such a solution. Class-imbalance techniques alter the distribution of training data. Operating thresholds select the threshold of probability at which positive predictions will be made. In our research, we have used the former, optimizing MCC on out-of-fold development predictions and maintaining the initial prevalence. Hence, we interpret the threshold performance comparisons as evidence of fixed decision rules and not evidence of synthetic balance of minority class.

4.3.2 MLP architecture ablation

The MLP dropout ablation involved a factorial combination of Dropout and Batch Normalization. Both are popular neural network building blocks, though the effectiveness of both is highly dependent on the network architecture and the dataset used. Using the selected configuration, adding Dropout resulted in an increase in the development ROC-AUC value of around 0.0036. On the other hand, using Batch Normalization decreased ROC-AUC by roughly 0.0047. This shows how a network optimization technique may be unhelpful to a specific problem.

The selected MLP maintained the Dropout rate of around 0.242 while discarding Batch Normalization. It has 44,801 parameters and was trained until 35 epochs using early stopping. These are the characteristics of the selected frozen model and should not be compared with another network.

The ablation presented here is crucial to evaluating how well MLP performs compared to gradient boosting. The neural network baseline did not serve as an untuned baseline and certainly did not represent some kind of canonical multilayer architecture. Instead, the parameters for regularization were experimented with on the dev set prior to the locked evaluation. Regardless, the magnitude of the ablation results depends on the choice of architecture, and does not mean that all possibilities for normalization and dropout have been explored.

SHAP was not part of the neural predictor or learning objective in this ablation study. This method was used for explanation post-training, and hence cannot be claimed as the reason for improvement in ROC-AUC score and decrease in loss.

4.3.3 Probability-calibration outcomes

Calibration methods were selected based on development out-of-fold probabilities prior to the opening of the final holdout fold. The calibration methods were different from model to model. Logistic Regression, XGBoost, MLP, and TabNet kept their probabilities uncalibrated. Random Forest, LightGBM, CatBoost, and FT-Transformer had sigmoid calibration. Isotonic calibration was used for RealTabR. The diversity was intentional since each frozen transform was the result of a choice made at the stage of development.

Sigmoid calibration of CatBoost led to small increases in performance in terms of Brier score, log loss, and ECE. The same is true for Random Forest in terms of probability quality. Sigmoid calibration of LightGBM led to improvements in Brier score and log loss, while leading to a slight increase in ECE. FT-Transformer also had improvements in log loss and ECE, but there was a small decrease in Brier score. These mixed results illustrate why calibration quality should be judged on multiple related measures rather than assumed from the selected transformation label alone (Niculescu-Mizil & Caruana, 2005; Guo et al., 2017).

The most obvious negative calibration consequence was RealTabR, in which the selected transformation based on development isotonic reduced holdout ROC-AUC by about 0.0007, PR-AUC by 0.0154, and increased log loss by approximately 0.0500. However, the transformation was left unchanged, as any post-holdout change would violate the evaluation lock established at the beginning of the study. This transforms the negative outcome into an indicator of limitations of calibration based on finite samples only, not as an incentive for optimizing the score afterwards.

Figure 3 displays calibration curves of the four selected models. Diagonal indicates correspondence between the mean predicted probability of churn and the frequency of events. Any visual closeness of the curve to the diagonal is to be assessed in parallel with the numerical Brier score, log loss, and ECE: visual alignment of the curve depends on the binning choice and may hide local differences of small observation groups.

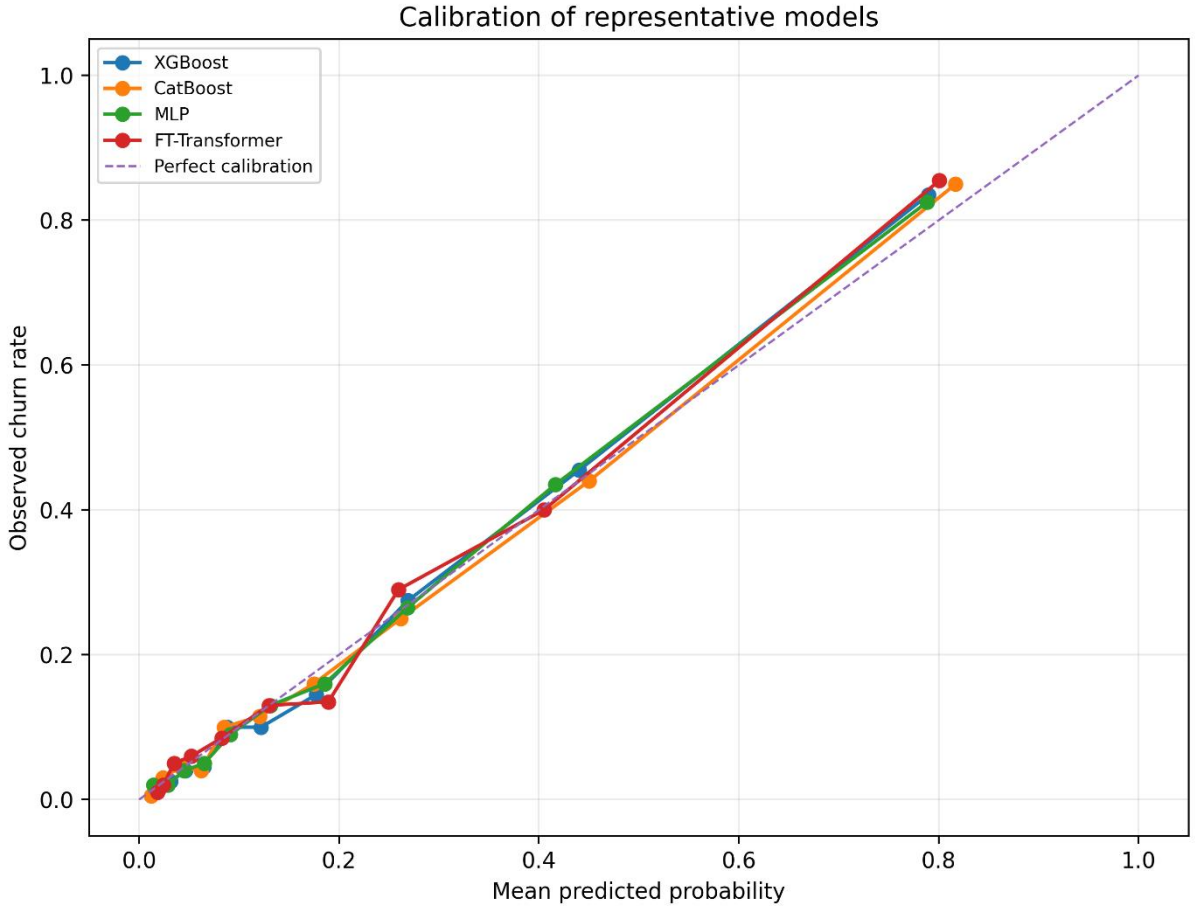


Figure 3. Calibration Curves of Representative Models

In summary, the calibration approach distinguishes between discrimination ability and probability accuracy. In addition, it sets the actual probability range in which future locking of decisions and post-hoc SHAP analysis will take place. The observed drawbacks associated with certain types of calibration are not eliminated in order to make the results look clean for the final model ranking.

4.4 Locked Holdout Performance

4.4.1 Discrimination and probability quality

Frozen probabilities were used to calculate a unique final probability vector for each of the nine models using the closed 2,000 observation holdout set. The results in Table 6 include ROC-AUC, PR-AUC, Brier score, F1, balanced accuracy, and MCC, all with 95% bootstrap confidence intervals. These confidence intervals are generated from 5,000 bootstrap samples of the fixed holdout data set, reflecting uncertainty conditional on the observed evaluation data set rather than an independent bank or time period.

CatBoost obtained the best holdout ROC-AUC performance with 0.8711 (95% CI [0.8507, 0.8898]). The second-best was XGBoost at 0.8707 [0.8494, 0.8896]. The best-performing tabular deep learning algorithm, FT-Transformer, scored 0.8685 [0.8474, 0.8873], and MLP attained 0.8669 [0.8454, 0.8860]. These point estimates indicate that there is little difference in pairwise ranking ability between these models compared with the performance of the Logistic Regression at 0.7762.

The positive class ranking and probability quality findings were not driven by the identical model as ROC-AUC. XGBoost produced the best PR-AUC score of 0.7223 (95% confidence interval: 0.6796 to 0.7612) and the lowest Brier score of 0.0986 (0.0900 to 0.1076). CatBoost was not far behind with PR-AUC of 0.7198 and Brier score of 0.0987 while FT-Transformer came in third with PR-AUC of 0.7177 and Brier score of 0.0993. The gap between the top two boosting algorithms is statistically trivial.

Table 6. Locked Holdout Performance with Bootstrap 95% Confidence Intervals

Model	ROC-AUC	PR-AUC	Brier score	F1-score	Balanced accuracy	MCC
CatBoost	0.8711 [0.8507, 0.8898]	0.7198 [0.6779, 0.7590]	0.0987 [0.0902, 0.1080]	0.6408 [0.6005, 0.6766]	0.7718 [0.7473, 0.7955]	0.5513 [0.5045, 0.5947]
XGBoost	0.8707 [0.8494, 0.8896]	0.7223 [0.6796, 0.7612]	0.0986 [0.0900, 0.1076]	0.6282 [0.5856, 0.6675]	0.7548 [0.7293, 0.7792]	0.5475 [0.4987, 0.5932]
FT-Transformer	0.8685 [0.8474, 0.8873]	0.7177 [0.6748, 0.7570]	0.0993 [0.0906, 0.1085]	0.6320 [0.5894, 0.6704]	0.7546 [0.7296, 0.7788]	0.5564 [0.5075, 0.6008]
MLP	0.8669 [0.8454, 0.8860]	0.7122 [0.6690, 0.7512]	0.1007 [0.0922, 0.1099]	0.6150 [0.5698, 0.6557]	0.7435 [0.7177, 0.7686]	0.5383 [0.4876, 0.5862]
LightGBM	0.8635 [0.8417, 0.8833]	0.7134 [0.6703, 0.7528]	0.1001 [0.0914, 0.1093]	0.6205 [0.5763, 0.6613]	0.7466 [0.7211, 0.7718]	0.5451 [0.4957, 0.5918]
Random Forest	0.8631 [0.8417, 0.8829]	0.7053 [0.6608, 0.7463]	0.1010 [0.0923, 0.1102]	0.6446 [0.6054, 0.6815]	0.7773 [0.7526, 0.8011]	0.5535 [0.5072, 0.5985]

Model	ROC-AUC	PR-AUC	Brier score	F1-score	Balanced accuracy	MCC
RealTabR	0.8615 [0.8395, 0.8815]	0.6976 [0.6534, 0.7389]	0.1004 [0.0911, 0.1100]	0.6028 [0.5552, 0.6469]	0.7239 [0.6989, 0.7489]	0.5652 [0.5166, 0.6103]
TabNet	0.8420 [0.8178, 0.8641]	0.6805 [0.6347, 0.7225]	0.1066 [0.0978, 0.1159]	0.5538 [0.5040, 0.6000]	0.6993 [0.6741, 0.7247]	0.5079 [0.4562, 0.5566]
Logistic Regression	0.7762 [0.7504, 0.8004]	0.4797 [0.4316, 0.5319]	0.1347 [0.1257, 0.1442]	0.4920 [0.4488, 0.5332]	0.6808 [0.6550, 0.7064]	0.3632 [0.3128, 0.4126]

Note. CI = percentile 95% confidence interval from 5,000 resamples of the same locked holdout observations. All F1, balanced accuracy, and MCC values use the model-specific classification threshold selected on development data.

The models LightGBM and Random Forest also maintained good predictive discrimination, with ROC-AUC equal to 0.8635 and 0.8631, respectively. The results were 0.8615 for RealTabR, 0.8420 for TabNet, and 0.7762 for Logistic Regression. Thus, the models based on gradient boosting continued to perform strongly when the assessment was done on hold-out data not seen before. On the other hand, the ranking of less well-separated models changed a bit in the hold-out sample, which is understandable in any finite hold-out scenario.

Figure 4 shows full threshold-independent ROC and Precision–Recall curves, providing graphical support for the numerical summary in Table 6. ROC curves demonstrate good discrimination power of the best models at different levels of false positive rate, and Precision–Recall curves reveal their ability to detect churners while producing as few false positives as possible. It should be stressed that the prevalence line is particularly important for PR analysis in this case as only 20.35% of the holdout clients are churners. An overall high accuracy can be achieved by predicting retention for everybody in the sample and is meaningless for retrieval of churners (Saito & Rehmsmeier, 2015).

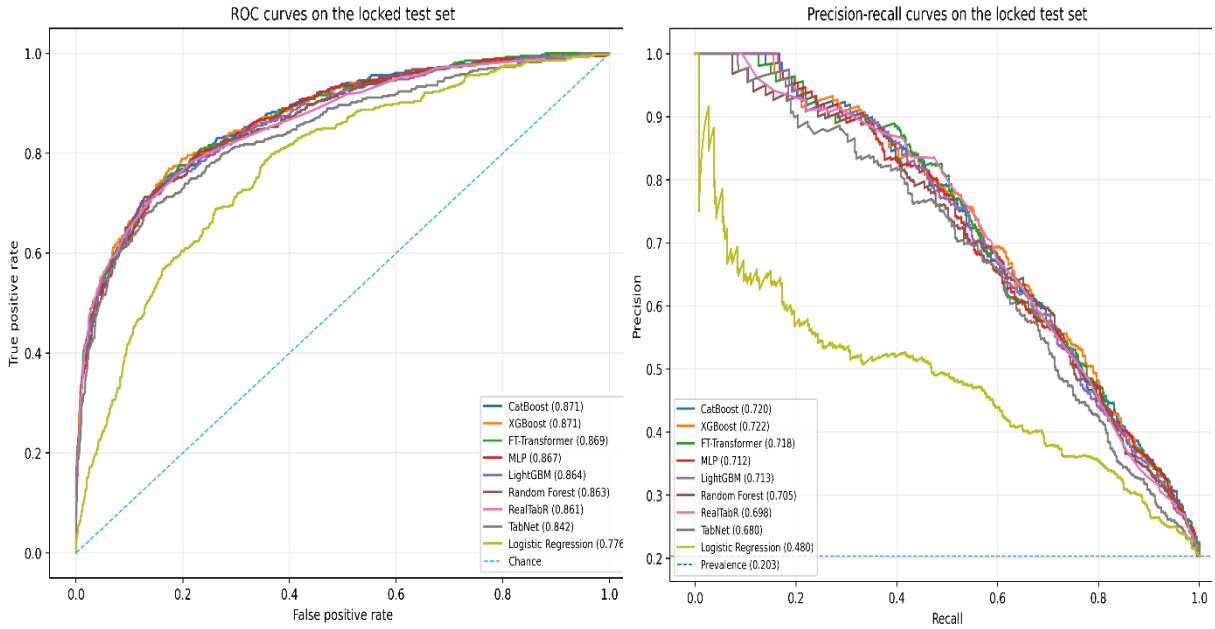


Figure 4. Locked Holdout ROC and Precision–Recall Curves

4.4.2 Locked-threshold classification and different operating preferences

The threshold-dependent metrics changed the ranking based on the point estimates. Random Forest showed the best F1-score (0.6446) and balanced accuracy (0.7773), while CatBoost demonstrated F1=0.6408 and balanced accuracy=0.7718. MCC was maximal for RealTabR (0.5652), then followed FT-Transformer (0.5564), Random Forest (0.5535), and CatBoost (0.5513). XGBoost reached MCC = 0.5475 and F1 = 0.6282. These metrics do not contradict the fact that either CatBoost or XGBoost could be leaders among the threshold-independent metrics, since the threshold selected defines the operating point.

In particular, MCC values were relatively high for RealTabR, which corresponded to a conservative decision strategy with precision equaling about 0.835 and recall 0.472. The model generated fewer false positives but missed more churners. The better F1 and balanced accuracy of Random Forest corresponded to another balance of recall and precision. A user focusing on minimizing unnecessary interventions would probably judge such decisions differently from a user trying to recover as many churners as possible.

However, the thresholds used in this benchmark are tuned to MCC based on development OOF predictions and thus are not optimal operational thresholds, which are usually set based on a bank's retention budget, contact costs, likelihood of treatment success and profit expectations. Thus, these thresholds allow for valid scientific comparison, but cannot be considered optimal operational thresholds for any real bank. In this regard, in Chapter 5 the discussion of the practical applications of the model will separately examine a capacity-constrained top-20% ranking process based on probabilities.

As for the top-20% ranking performance measure, the top-ranked models were able to capture about 62%-63% of actual churn customers among those 20%. This demonstrates a concentration of the churn cases observed and a potential case for prioritization. However, this doesn't mean that contacting these customers will prevent these churns because a separate assessment of the effectiveness of such retention activity is needed.

4.4.3 Holdout uncertainty and the limits of point estimates

In Figure 5, the ROC-AUC holdout estimate point estimates and their model-level bootstrap confidence intervals are compared. The confidence intervals of CatBoost, XGBoost, FT-Transformer, and MLP overlap almost entirely. This chart conveys that the point estimates are uncertain, but does not indicate a p-value of the difference of point estimates. Since both models predict on the same customer, the variance of interest in comparing any two models is the covariance of their predictions.

The model-level interval cannot prove equivalence since they are overlapping. For instance, the interval of CatBoost cannot capture the sampling distribution of CatBoost - XGBoost because the models may be correlated in making their mistakes. In that case, the paired test described in Section 4.5 makes the more rigorous test of small differences

between models (DeLong et al., 1988). This is one of the main reasons why ranking the models according to their accuracy in four decimal places is meaningless.

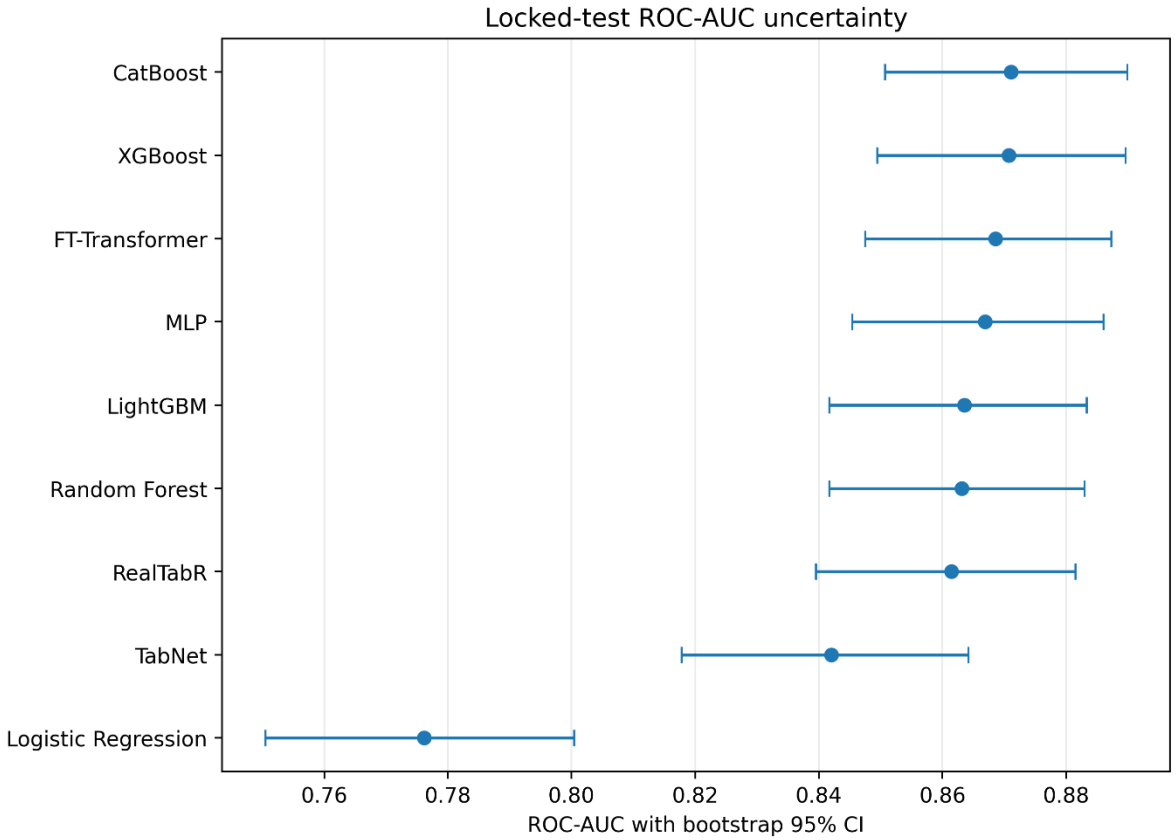


Figure 5. ROC-AUC Estimates with Bootstrap Confidence Intervals

4.5 Statistical Comparison of Predictive Models

4.5.1 Paired ROC-AUC tests and multiplicity correction

Pairwise inferential procedures were applied to holdout predictions that cannot be changed further. Table 7 contains DeLong test results for each other model against XGBoost, taken to be the reference model in this case. Difference columns contain values of ROC-AUC of the comparator model subtracted from ROC-AUC of XGBoost. Positive difference values mean superior results of the comparator; negative values indicate the superiority of XGBoost. Holm-adjusted p-values were determined within the family of ROC-AUC comparison. The multiplicity was considered as of nine-model benchmark rather than of two models shown in comparisons (Holm, 1979).

The CatBoost model had the estimated advantage of 0.0003 in the underlying paired comparison, with 95% CI [−0.0038, 0.0044], and p = 1.0000 (adjusted). FT-Transformer and MLP models had difference values of −0.0022 and −0.0039 with adjusted p = 1.0000 in both comparisons. None of the three comparisons turned out statistically significant after correction. In practical terms, the data failed to show any significant

ROC-AUC difference between XGBoost and the alternative models under consideration.

Table 7. Paired DeLong ROC-AUC Comparisons Against XGBoost

Comparator	ROC-AUC difference	95% CI	Holm-adjusted (p)	Significant after Holm correction
CatBoost	+0.0003	[−0.0038, 0.0044]	1.0000	No
FT-Transformer	−0.0022	[−0.0093, 0.0048]	1.0000	No
MLP	−0.0039	[−0.0110, 0.0032]	1.0000	No
LightGBM	−0.0072	[−0.0104, −0.0041]	0.0001	Yes
Random Forest	−0.0076	[−0.0132, −0.0021]	0.1356	No
RealTabR	−0.0093	[−0.0173, −0.0012]	0.3824	No
TabNet	−0.0287	[−0.0410, −0.0165]	0.0001	Yes
Logistic Regression	−0.0946	[−0.1167, −0.0725]	<0.0001	Yes

Note. Difference = comparator ROC-AUC minus XGBoost ROC-AUC. Adjusted p-values use Holm correction within the nine-model ROC-AUC comparison family. The comparison signs are based on underlying non-rounded estimates.

Conversely, the contrasts in the case of LightGBM (−0.0072; adjusted p = 0.0001), TabNet (−0.0287; adjusted p = 0.0001), and Logistic Regression (−0.0946; adjusted p < 0.0001) proved to be significant post-Holm adjustment. On the other hand, the unadjusted confidence intervals for Random Forest and RealTabR were below zero, but their adjusted p-values equaled 0.1356 and 0.3824, respectively. Thus, in these two cases, the contrasts did not prove to be significant using the familywise criterion. In both cases, reporting the adjusted p-value prevents any biased representation of a comparison.

The puzzling result of this comparison is the significance of the contrast between XGBoost and LightGBM despite the intersection of the intervals of the corresponding AUCs. Such an outcome can be explained by the paired design and not inconsistency of the statistical calculations. It is important that the models in question are evaluated based on the same sample, and their AUC score correlation impacts the variance of their performance difference. While separate intervals show uncertainty about each AUC level, the paired DeLong method compares two correlated AUCs (DeLong et al., 1988).

The non-statistically significant results for CatBoost, FT-Transformer, and MLP should not be interpreted as an indicator of equality among the four features. The inability to reject the difference hypothesis implies that there was no significant difference found at the adjusted criterion. Neither does it set any pre-defined equivalence boundary nor can it claim the same ranking for a different bank or time period.

4.5.2 Paired comparisons across other metrics

The numerical-statistical difference between ROC-AUC extended to other metrics. Specifically, XGBoost was significantly better than RealTabR, TabNet, and Logistic

Regression for the PR-AUC value after Holm correction. The non-significant comparisons between the PR-AUC values of XGBoost and LightGBM and Random Forest could not be adjusted for multiplicity. As a result, the best point estimate of XGBoost's PR-AUC suggests a promising ordering of positive classes but does not imply that XGBoost significantly outperformed all competitors in this case.

XGBoost was significantly different from TabNet and Logistic Regression but not from other leading classifiers for the Brier score and F1-score. Finally, Logistic Regression was the only classifier significantly inferior to XGBoost in MCC after adjusting for multiplicity. Similarly, exact McNemar tests on fixed binary decisions revealed that Logistic Regression was the only classifier whose classification accuracy significantly differed from XGBoost after Holm correction. It becomes clear from the results that the identity of a statistically significant comparison varies depending on the metric under analysis, just like descriptive rankings.

Of the 36 possible paired comparisons, significant adjusted outcomes clustered primarily among those including Logistic Regression or TabNet. Strong boosting and deep tabular models were hard to differentiate significantly. This limits the primary research conclusion, which does not demonstrate systematic superior performance of deep models as compared to tuned gradient-boosting, nor can it be used to confirm that each paired combination of best models is fully interchangeable.

Paired comparison improves the benchmark through its recognition of repeated observations of the same customers. It cannot fix the lack of an independent entity and temporal validation set, nor can it make a single public data set representatively sampled for all banking demographics. Further research on additional aspects of model quality follows below without implying any of these as the proof of generalization.

4.6 Cross-Model Explainability Results

4.6.1 Attribution fidelity and four-model consensus

Comparisons of SHAP values could only be done once all issues with predicting artifacts were fixed. XGBoost, CatBoost, MLP, and FT-Transformer pipelines restored the probabilities stored in archives to 1.0 correlation – the maximum numerical difference being zero for both boosting models and about 2×10^{-7} for the neural networks. Permutation-SHAP additive reconstruction tests also confirmed the agreement with modeled calibrated probabilities up to floating-point error. That is important as an explanation provided for a different model does not necessarily reflect the original model.

Four representative architectures were explained within the same ten features input space based on the same 300 customers stratified holdout sample and the same 100 customers development sample. Global importance was computed as an average absolute attribution in probability scale and normalized per each model prior to computing cross-model average. It prevents native tree-split importance and deep-model explanation scale from affecting the consensus due to the different units of measurement.

Table 8 describes the outcome consensus. Age came out first with average normalized importance of 0.2999, followed by NumOfProducts with average normalized importance of 0.2627, while IsActiveMember took the third place with average normalized importance of 0.1471. The cumulative importance of the three variables constituted around 0.7097, which was roughly 71.0% of all normalized importance. All three were found in the top three importance ranks in all of the four exemplary models.

Table 8. Cross-Model SHAP Consensus Importance

Consensus rank	Feature	Mean normalized importance	SD across models	Mean model rank	Models including feature in Top 3	Models including feature in Top 5
1	Age	0.2999	0.0233	1.00	4	4
2	NumOfProducts	0.2627	0.0102	2.00	4	4
3	IsActiveMember	0.1471	0.0108	3.00	4	4
4	Geography	0.0893	0.0190	4.25	0	4
5	Balance	0.0699	0.0155	5.25	0	2
6	Gender	0.0685	0.0051	5.50	0	2
7	CreditScore	0.0219	0.0055	7.50	0	0
8	EstimatedSalary	0.0207	0.0047	7.75	0	0
9	Tenure	0.0111	0.0033	9.25	0	0
10	HasCrCard	0.0089	0.0058	9.50	0	0

Note. Importance is normalized separately within each model before averaging across XGBoost, CatBoost, MLP, and FT-Transformer. Top-k counts refer to the four representative models.

The variable "Geography" was fourth most important (0.0893), "Balance" was fifth (0.0699), and "Gender" was sixth (0.0685). There was some variability in the order of these mid-rank variables; "Geography" was in the top five of all four models, while "Balance" and "Gender" were in only two top fives each. "CreditScore," "EstimatedSalary," "Tenure," and "HasCrCard" together had very low normalized contributions. An importance value near zero does not mean that the input is unimportant in every single prediction, but it means that the mean absolute contribution is small in this explanation framework.

Statistics about agreement quantified this apparent similarity. Pairwise Spearman rank correlations for the four models were from 0.927 to 0.988, and Kendall correlations were from 0.822 to 0.956. Each pair had its three highest-ranked features in common; four of the six pairs had four features in common among their top five, and XGBoost and FT-Transformer had five features in common. Thus, the top three finding was not simply because the boosting model was compared to another boosting model, as there were also feed-forward networks and attention models included.

Bootstrapping the explanation cohort offered an independent check on sampling stability. Across 2,000 bootstraps, mean rank correlation between the importance

orderings of bootstrapped and full cohorts was found to range between 0.993 and 0.996. The three most significant features were always in the top three of all four models in each bootstrap replicate. The former tests whether different models give similar emphasis to the same input features, while the latter tests the robustness of importance ranking in a single architecture to the particular set of samples used.

Figure 6 illustrates the consensus by a bar plot with standard deviations across all models. The main take-away is the concentration of explanation mass across three input features and the low variability of this order within architectures. None of both the bar lengths or agreement coefficients assess any causal relationships between input and output variables. They simply estimate the way in which the already fitted prediction function assigns attribution weights in the defined background and holdout samples (Lundberg & Lee, 2017; Aas et al., 2021).

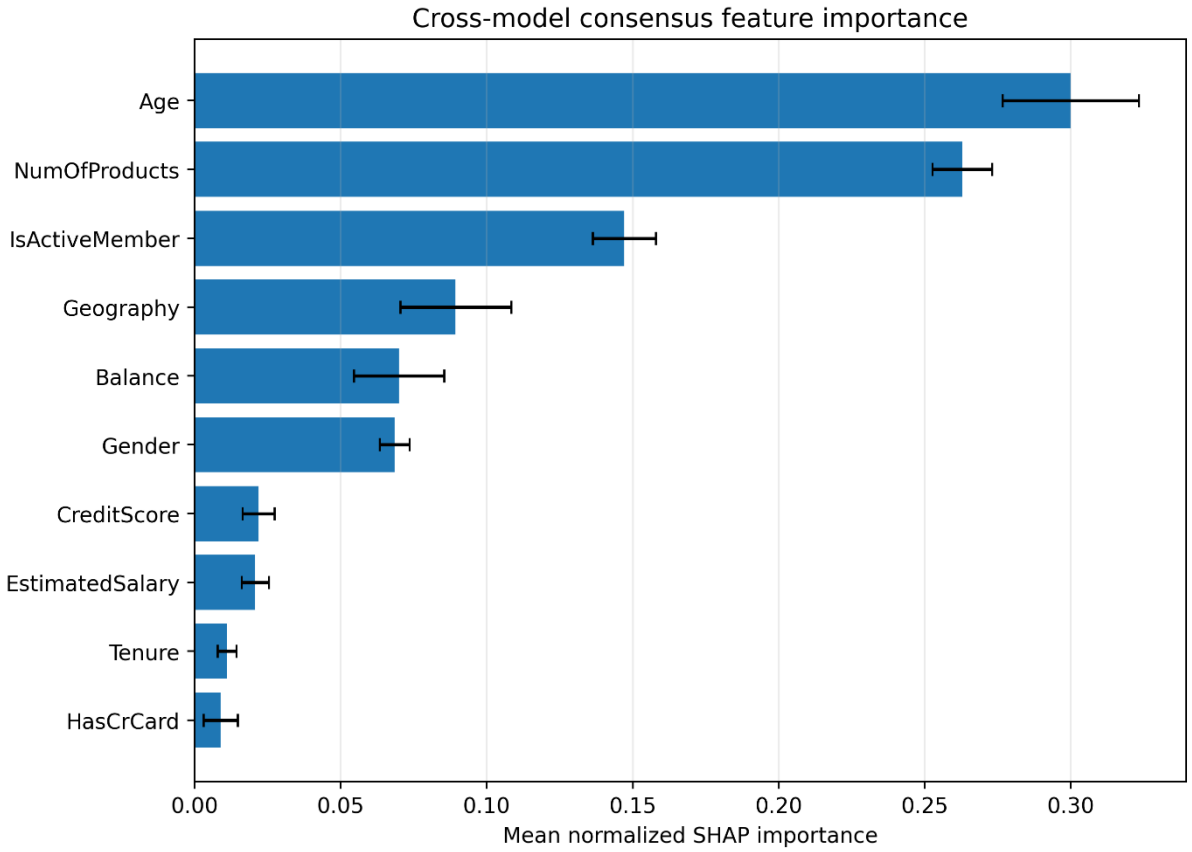


Figure 6. Cross-Model SHAP Consensus Importance

4.6.2 Direction and nonlinear shape of feature contributions

The global mean absolute SHAP importance gives an indication of how much each feature contributes on average, but not whether high or low values contribute to the likelihood of churn. The signed profiles were therefore considered in order to see which directions had been learnt. For age, there was an overall increasing trend, meaning that the contributions were usually negative for ages close to 18-39 years old, started becoming positive around 40-45 years old and then became very large for people aged

46 or more. This is a trend that held true across the four representative models, although its exact form depends on the observed population and fitting functions used.

There was a binary pattern of activity status. The inactive group contributed, on average, positively in the order of +0.051, while the active group had a negative contribution of about -0.049. In other words, activity status was always one of the three most important features, and, furthermore, the signed feature showed that being an active member was associated with less churn in the fitted model. It is worth mentioning that the claim above holds for the fitted prediction function and not the churn per se.

NumOfProducts showed a nonmonotonic relationship. One-product customers had a positive contribution, two-product customers had a negative contribution, while three-product customers had a very strong positive contribution. Nevertheless, this was a conclusion made on the basis of only seven three-product observations in the explanation sample. This conclusion should not be generalized into a general rule and, especially, does not agree with the simplistic monotonic claim that increasing the number of products decreases churn.

Balance, EstimatedSalary, and Tenure were also found to have nonmonotonic signed importance profiles, though the relatively lesser importance value suggests they were less impactful on the overall probability explained variation. Higher CreditScore is associated with lower predicted churn, although it is ranked seventh in importance. In terms of categorical variable, Germany had positive mean contribution while France and Spain had negative contributions. Observations in the category Female had positive average contribution compared to the employed explanation reference, while observations in the category Male had negative contributions.

This refers to the association between input values and the attributions from the model, which do not translate into an operational intervention effect. It relies on the reason why the groupings differ, whether they are suitable for use in a particular jurisdiction, and whether a retention effort impacts churn outcomes. The subgroup analysis and direct sensitivity analysis of Section 4.8 explore these aspects without interpreting the attribution as policy guidance.

4.6.3 TabNet explanation agreement

Furthermore, the explanation protocol was applied to the final TabNet estimator to investigate whether the main attribution pattern can be found in a model with reduced predictive discrimination capacity. The serialization of its estimator and preprocessing pipeline was successfully reconstructed. Compared to the holdout immutable raw scores, the maximal absolute probability difference was 8.79×10^{-7}, the prediction correlation was 1.0000, and the set predefined fidelity tolerance of 10^{-5} was met. The maximal residual of SHAP additive reconstruction was 1.48×10^{-7}, and calculation took 93.44 seconds.

All ten TabNet attributions with their importance are provided in Table 9. The top-three variables by importance were NumOfProducts (0.307140), Age (0.289845), and IsActiveMember (0.137351), while Geography and Gender came in fourth and fifth

places, correspondingly. Despite the fact that this ranking does not repeat the consensus top-three pattern, it includes the very same three leading features. It is worth mentioning that the holdout ROC-AUC of TabNet was noticeably inferior compared to the four key explanatory models. It implies that the main predictive factors obtained from this dataset were not limited to the best classifiers.

Table 9. Permutation-SHAP Feature Importance for the Final TabNet Model

Rank	Feature	Mean absolute SHAP	Normalized importance	Four-model consensus rank
1	NumOfProducts	0.100076	0.307140	2
2	Age	0.094441	0.289845	1
3	IsActiveMember	0.044753	0.137351	3
4	Geography	0.037931	0.116412	4
5	Gender	0.021095	0.064744	6
6	Balance	0.010956	0.033626	5
7	EstimatedSalary	0.005618	0.017244	8
8	CreditScore	0.005566	0.017084	7
9	HasCrCard	0.003692	0.011331	10
10	Tenure	0.001702	0.005225	9

Note. TabNet is explained on the same stratified 300-customer cohort and with the same 100-customer development background used for the four-model consensus. Values are from the frozen restored estimator.

The results from the comparison using extended rankings have been shown in Table 10. The Spearman correlation of TabNet with respect to the consensus ranking has been 0.9515 ($p = 0.000023$), where there is full top three ranking coincidence (3/3) and four out of five consensus-ranked leading features preserved in their top five ranking (4/5). It thus confirms the key explanation pattern and shows some divergence in terms of the ranking of Geography, Gender, and Balance features.

Table 10. TabNet Agreement with the Four-Model SHAP Consensus

Model	Spearman ρ	p-value	Top-three overlap	Top-five overlap	Top three TabNet features
TabNet	0.9515	0.000023	3/3 (100%)	4/5 (80%)	NumOfProducts, Age, IsActiveMember

The TabNet evidence represents a further validation of explanation consistency rather than a ninth model's explanation certificate. The other benchmarked models did not go through the same reconstruction validated SHAP process, and one of them - RealTabR – faced a fidelity problem. It is important to hold both the explanation population and

the algorithm constant among the restored models to avoid unnecessary methodological differences.

4.6.4 RealTabR explanation-fidelity limitation

This last fitted RealTabR model was impossible to serialize, as the library-implementation had a local lambda function inside. This attempt at recreation was only made to see if it would yield valid attributions for the locked final estimator, using the stored hyperparameters, the initial training manifest, a fixed random seed 2027, and the exact same 6800/1200 train/early stopping split.

These reconstructed scores did not satisfy the demanded fidelity tolerance. Their maximum absolute error from the archived probabilities was 0.2373, their mean absolute error was 0.0341, and their correlation was 0.9909. It was clear that this correlation value was not sufficient evidence that these two functions gave the same numerical predictions, considering that a single error can be enough to change the explanation per customer.

The SHAP values derived using this reconstruction have been disregarded accordingly. The reporting of such data under the heading of the original RealTabR estimator would incorrectly represent the system used to generate the results of this benchmark. Such restrictions do not discredit the archived ROC-AUC, PR-AUC, or threshold scores of the RealTabR estimator, as these are still calculated using the original fixed predictions. The only effect is to limit the applicability of cross-model explainability.

4.7 Robustness Under Input Perturbations

4.7.1 Numerical noise and categorical corruption

Predictive accuracy on a pristine holdout does not imply that the model is immune to certain types of measurement errors. Hence, the test involved quantifying changes from the fixed baseline predictions under seven stress tests that occurred 20 times each. These stress tests included minor and moderate numeric perturbations, 5% and 10% missing data, 5% and 10% categorical perturbations, and combined stress test. The four models were assessed using the initial parameters, calibrators, and thresholds for the models.

Minor numerical perturbations resulted in a loss of 0.001 in the ROC-AUC score for XGBoost, CatBoost, and MLP. In the three models, the Spearman correlation coefficient between perturbed and baseline probability rank scores was higher than 0.993, while fewer than 1.6% of the locked classes changed.

The FT-Transformer model was more vulnerable to that very same mild scenario. Its ROC-AUC decreased by approximately 0.0106, its rank correlation became 0.895, and 2.9% of classifications were modified. Under moderate numeric noise, the three other representative models lost 0.0064 – 0.0070 ROC-AUC points, while FT-Transformer lost roughly 0.0189. As such, neural networks with comparable discrimination performance in clean data may have different vulnerability levels to changes in input.

The substitution of categorical values created less impact than missingness simulation. Changing 10% of recorded Gender and Geography values into alternative valid

categories caused an approximate ROC-AUC reduction of 0.0050 – 0.0060, along with a modification of 1.1% – 1.5% of fixed classifications. Even though categorical noise had relatively low cumulative impact in that particular testing scenario, it is still significant for the governance analysis, as these features are also analyzed regarding sub-group behaviors and probability sensitivity.

4.7.2 Missingness and combined input stress

Masking the recorded fields resulted in the most pronounced degradation. Using median/mode values from development data to replace 10% of input cells decreased ROC-AUC by about 0.0242 for XGBoost, 0.0241 for CatBoost, 0.0249 for MLP, and 0.0259 for FT-Transformer. In this case, PR-AUC dropped by about 0.046 – 0.050. The effect size is significant compared to very small differences in clean-data ROC-AUC between state-of-the-art models.

Masking does not produce a realistic missing data generation mechanism seen in banks. Instead, this method imitates the partial loss of original data and its deterministic substitution, hence it can estimate sensitivity only in this particular situation. First of all, model performance depends on the set of masked features, values that were used for replacement, and missingness probability for different types of customers. This effect has not been confirmed elsewhere yet.

When subjected to both types of stresses, XGBoost, CatBoost, and MLP lost 0.020 ROC-AUC, while FT-Transformer lost about 0.031 ROC-AUC with the biggest probability shift and binary flip-rate. However, this sequence was not universal: XGBoost demonstrated good stability under the masking and combined stress; meanwhile, MLP was particularly resistant to numeric perturbation. These nuances make it impossible to say which is the best-performing model overall based only on the single metric.

Fig. 7 shows the ROC-AUC shifts for the given stress scenarios. If the point is close to 0, it means that the degradation is low compared to the corresponding un-perturbed baseline. This graph has to be considered in conjunction with the correlation of probability ranks and binary flip rates of the models. Two models may have equal ROC-AUC shift but still differ in terms of how they change the set of customers identified by the fixed threshold.

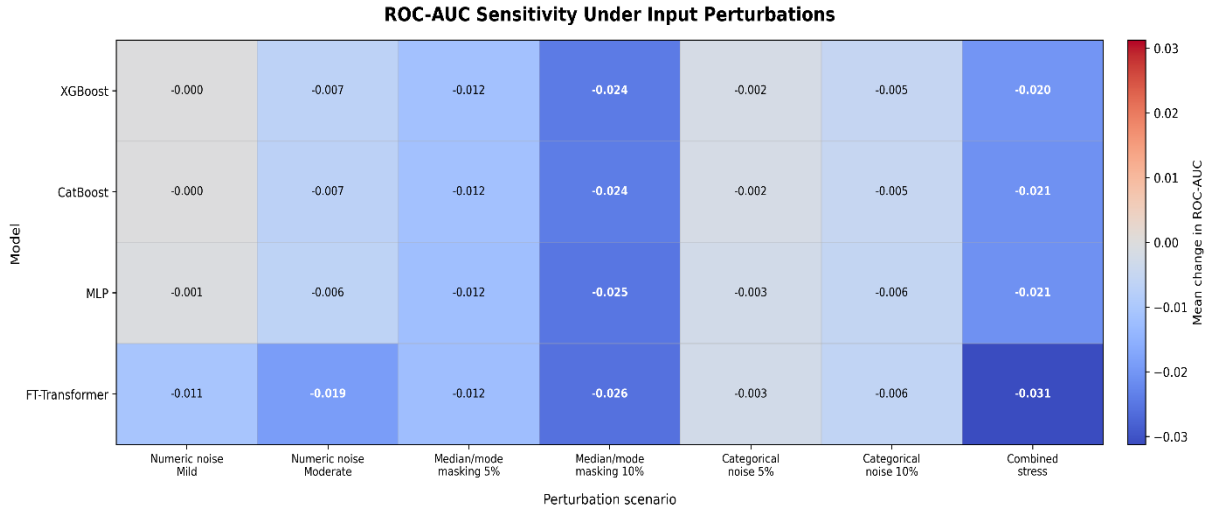


Figure 7. Robustness Under Controlled Input Perturbations

The key empirical takeaway is that even an excellent predictor based on clean data could still be exposed when original field values are not available. A mere difference of several thousandths in ROC-AUC of clean data would not automatically make up for the drop of 0.024–0.026 in the context of masking. However, this experiment does not pretend to prove the resistance against natural temporal drift, new banks, malicious behavior, or all possible data perturbations. The main message of the experiment is related to the robustness to the particular simulation of the frozen holdout.

4.8 Subgroup Performance and Direct Sensitivity

4.8.1 Composition and observed churn prevalence

Subgroup analysis has been done on 2,000 locked holdout customers by using the current probabilities and classification decisions. Variables used for subgrouping are Gender, Geography, Age, and IsActiveMember. The first step has been to analyze the churn rate of each subgroup rather than analyzing any differences between positive predicted rates alone. Differences between baseline churn rates influence precision, probability-quality metrics, and some fairness metrics even if the levels of discrimination are broadly similar (Hardt et al., 2016; Pleiss et al., 2017).

Holdout contains 930 Female customers and 1,070 Male customers. Churn rate in the Female subgroup is 25.05%, while churn rate in the Male subgroup is 16.26%. Churn rates in different geographical regions are 17.14% in France, 29.96% in Germany, and 17.05% in Spain. So, it is clear that the churn rate in the German subgroup is significantly higher than the two other subgroups.

The disparities between age groups were more pronounced in the predefined age intervals. There was 6.72% churn rate in the 18-34 age group, 18.56% in the 35-44 age group, 46.75% in the 45-54 age group, and 40.12% in the age group above 55 years old. The inactive customers have a prevalence of 27.48% as opposed to the 13.66% for the active customers.

4.8.2 Gender-related discrimination and error-rate disparities

The predicted positive rate was higher in Female customer subgroup than in the corresponding Male subgroup across all nine models. For the best-performing models, the Female-minus-Male differences varied between approximately 0.071 and 0.102. These differences cannot be directly interpreted as discrimination since the difference in the target prevalence was higher for females. Evaluation of the error rates and probability accuracy offers a more thorough explanation of model behavior.

The ROC-AUC differences between Female and Male subgroups were less than 0.01 in most cases among the best-performing nonlinear models, which suggests a similar ranking capability. However, the corresponding threshold decisions had higher false-positive rates for Females. The differences in False Positive Rates between Females and Males were 0.034 for XGBoost, 0.050 for CatBoost, 0.042 for MLP, and 0.044 for FT-Transformer. Bootstrap intervals for these differences were above zero.

All nine methods also achieved greater Brier scores for females, meaning that their probability prediction was less accurate for females than for males according to the tested distributions. Recall differences were smaller overall, and their 95% confidence intervals typically contained zero for the best nonlinear methods. Logistic Regression showed considerably larger gender differences on both recall and FPR. This contrast highlights the value of evaluating both threshold-independent discrimination and threshold-dependent errors: equal ROC-AUCs do not imply equal performance at an operating point.

4.8.3 Geography, age, and activity status

Regarding geographical categories, Germany always scored the highest in terms of positive-predictive value, recall, false-positive rate, and Brier score. In terms of XGBoost, the ranges in recall of the geography groups were 0.217, whereas in terms of FPR, the ranges were 0.093. High prevalence of churn among the German population is an important point to consider regarding these discrepancies, yet the presence of high prevalence does not necessarily explain why all gaps between the operating points occur.

The greatest difference occurred among age groups. In terms of XGBoost, the range in positive-predictive value for age bands was 0.576, the range in recall was 0.754, and the range in FPR was 0.348. Overall, it can be seen that the 18-34 age group had the lowest recall, while the 45-54 age groups had the highest recall and FPR. These gaps are consistent with the results from the SHAP analysis of the strong age dependency; nevertheless, the SHAP profile on its own cannot show whether the obtained errors are acceptable or not.

In addition to this, activity status modified the decision behavior. Inactive users received a higher risk score and had a higher recall rate and FPR. In the case of the XGBoost model, the recall difference related to activity was equal to 0.196, and FPR difference was 0.051. Hence, the same variable with the large SHAP contribution had a significantly different decision behavior when considering the threshold. This is an

indicator of dependency of the score and the decision behavior of the fitted model on engagement, not on the causal effect of recording activity on churn reduction.

The observed differences in groups depend on the fixed common thresholds; hence, there is no need to consider subgroup-specific thresholds based on hold-out. These observed differences describe the initial evaluation system. Different threshold strategy will modify recall and FPR difference; however, for analysis and recommendations, it will be required to have the objective evidence not provided by the benchmark.

4.8.4 Direct probability sensitivity

The complementary diagnostic varied one audited categorical predictor at a time while holding the other nine predictors entered constant. Specifically, each record was assessed in both cases of being assigned either Female or Male for Gender and being assigned to France, Germany, and Spain for Geography. This is a method to measure the direct dependence of the fitted probability function on the audited input while holding the rest of the entered inputs constant; it does not provide a causal intervention regarding that person's demographic characteristic.

Assigning Female rather than Male results in an increase in the mean churn probability by 0.0445 for XGBoost, 0.0446 for CatBoost, 0.0535 for MLP, and 0.0436 for FT-Transformer. The confidence interval for the difference in means was above zero. Depending on the model, 94.8% to 100% of holdout records were predicted to have higher probability of churn when Female compared to Male. However, the binary locked classification decision changes in only about 4.1% to 5.6% of the records.

Changes due to geography had a bigger effect on each customer. In three different geographic groupings, the mean probability ranges were 0.083 for XGBoost, 0.088 for CatBoost, 0.129 for MLP, and 0.103 for FT-Transformer. Around 10.4%–17.6% of cases saw a shift in locked classification by varying only geography. MLP was the most sensitive to changes in geography among the models compared here.

The Germany groupings always yielded higher churn probabilities than those of France and Spain. The mean differences between Germany and France were around 0.068–0.116, while the mean Germany minus Spain difference was 0.059–0.107. Figure 8 is the combined view of gender and geography comparisons, demonstrating that both the predicted probability and the decision threshold are dependent on sensitive information.

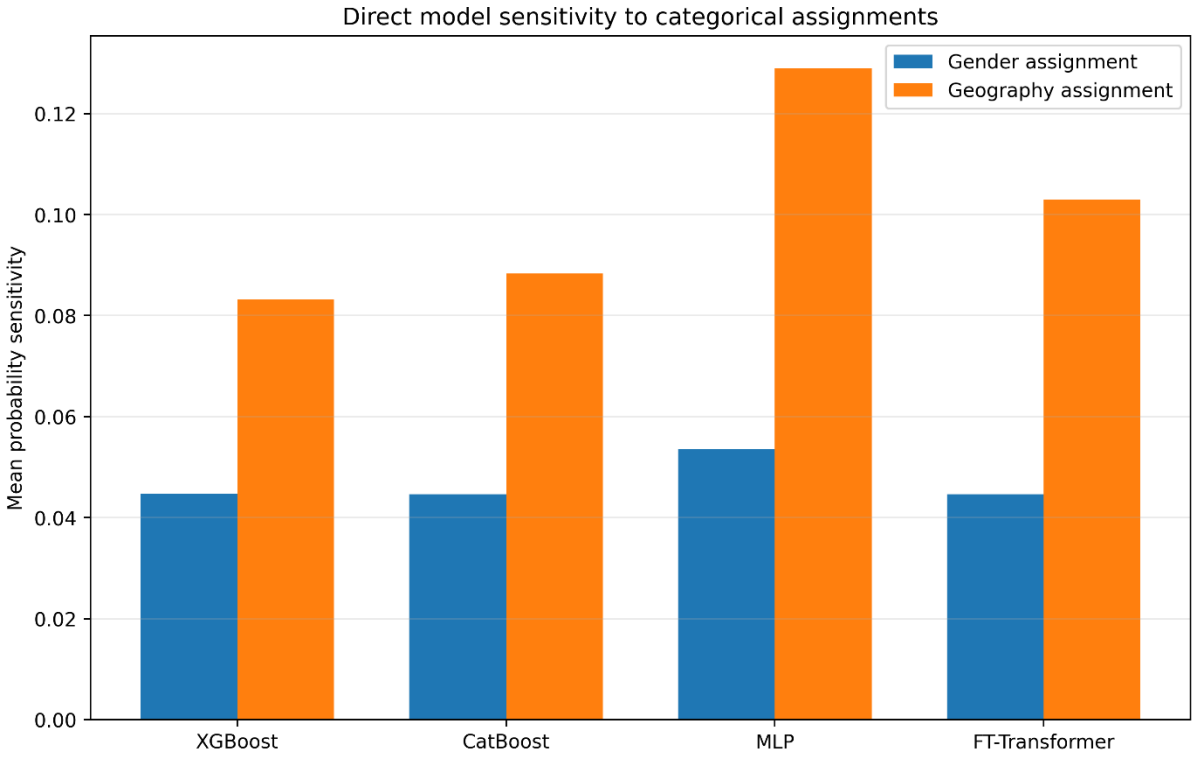


Figure 8. Direct Sensitivity to Gender and Geography Assignments

Direct sensitivity is a diagnostic tool, but not a measure of legal fairness, proof of counterfactual fairness, or causal effect. Swapping one feature may result in combinations of features that do not occur in the joint observed distribution. Moreover, legal and ethical analysis is dependent on an institutional setting, application purpose, relevant rules, and intervention effects. Thus, the observed group and sensitivity with respect to the input features are reasons for close monitoring within the boundaries of the available data (Hardt et al., 2016; Pleiss et al., 2017).

4.9 Computational Efficiency and Memory Assessment

4.9.1 Predictive performance and execution cost

Usefulness of the comparative models also hinges on the amount of resources needed for training, storing, explaining, and running the predictor. All timing measurements have been performed using only a CPU-only Kaggle setup with four cores, and the final classical classifiers have been fitted to the entire set of 8,000 development samples while the final deep models have been fitted according to their 6,800 gradients + 1,200 early stopping procedure. The values of the immutable locked ROC-AUC along with the final fitting time, inference latency, and the stored artifact size have been collected into Table 11.

The XGBoost model achieved ROC-AUC 0.8707 with final fitting time equal to 0.132 seconds, inference latency 0.0043 milliseconds per customer, and artifact size of 0.276 MB. CatBoost had slightly better ROC-AUC of 0.8711 but needed 2.391 seconds of fitting, that is about 18.1 times more than XGBoost in the recorded environment.

Inference latency of CatBoost was 0.0022 milliseconds per customer and its artifact size was 0.213 MB. Therefore, XGBoost was not unique in being the fastest model on all the computational metrics.

Table 11. Locked Predictive Performance and Computational Efficiency

Model	ROC-AUC	Final fit time (s)	Fit time relative to XGBoost	Inference latency (ms/customer)	Artifact size (MB)
CatBoost	0.8711	2.391	18.1×	0.0022	0.213
XGBoost	0.8707	0.132	1.0×	0.0043	0.276
FT-Transformer	0.8685	53.319	403.7×	0.1014	8.351
MLP	0.8669	7.023	53.2×	0.0013	0.175
LightGBM	0.8635	0.574	4.3×	0.0153	1.640
Random Forest	0.8631	2.077	15.7×	0.0722	16.662
RealTabR	0.8615	127.194	963.1×	0.1601	Not available
TabNet	0.8420	82.456	624.3×	0.0725	Not available
Logistic Regression	0.7762	0.032	0.2×	0.0026	0.004

Note. Fit and inference times correspond to the original locked evaluation in a four-core CPU environment. Inference timing was reported per customer. Artifact sizes are unavailable for TabNet and RealTabR in the original final comparison; do not infer stored deployable files.

Of the deep architectures, MLP had the lowest inference latency at 0.0013 milliseconds per customer and a relatively small 0.175 MB artifact, but needed 7.023 seconds for training, 53.2 times the XGBoost fit time. FT-Transformer was close in ROC-AUC performance with 0.8685 but needed 53.319 seconds for training (403.7 times XGBoost), approximately 0.1014 milliseconds per customer during inference, and 8.351 MB artifact size. Permutation-SHAP explanation of the model took 363 seconds, or 11.6 times the explanation time of the default XGBoost model.

TabNet needed 82.456 seconds for training, while RealTabR needed 127.194 seconds, which translates to 624.3 and 963.1 times the fit time of XGBoost, respectively. Both ROC-AUC scores for the two models were lower compared to that of the leading boosting models. The Random Forest fit time was 2.077 seconds, while the stored artifact was relatively large at 16.662 MB, given its 450 trees and over 216,000 nodes. LightGBM took 0.574 seconds to train, faster than CatBoost but slower than XGBoost in this comparison.

These latency numbers should be interpreted carefully since some holdout forecasts completed in mere milliseconds. Smaller latencies measured per customer are affected by constant overhead costs, batching, process state, and reporting accuracy. These are not measures of throughputs under actual concurrency, network costs, authentication, and serving API. Differences in fit time are more meaningful in this controlled environment but may differ in other CPU and library environments.

4.9.2 Post-test process-memory measurements

Memory peak consumption was determined through isolated post-test replications of the already-frozen model configurations. In this test, process-tree resident set size (RSS) sampling occurred at around 2 ms intervals within separate processes, measuring both a process baseline and a memory peak for training and inference phases. As opposed to the original predictor, the memory analysis retrained the frozen configurations for instrumenting purposes. The purpose of this analysis is to describe the computational footprint and not substitute the final archived predictor or prediction vector.

Table 12 retains the six columns of memory data: training baseline RSS, training peak RSS, incremental training peak, inference baseline RSS, inference peak RSS, and incremental inference peak. It is necessary to distinguish between the absolute and incremental memory values since a high baseline RSS value for a model may indicate the memory allocated by libraries and interpreters and not by parameter fitting.

Table 12. Post-Test Peak-Memory Measurements for Frozen Configurations

Model	Training baseline RSS (MB)	Peak training RSS (MB)	Incremental training peak (MB)	Inference baseline RSS (MB)	Peak inference RSS (MB)	Incremental inference peak (MB)
Logistic Regression	187.33	192.60	5.27	188.27	188.81	0.55
Random Forest	195.21	219.15	23.94	217.29	217.64	0.35
XGBoost	211.94	220.53	8.59	218.97	219.16	0.19
LightGBM	271.27	281.52	10.25	275.27	275.28	0.01
CatBoost	261.78	280.08	18.30	280.08	280.52	0.44
MLP	423.51	448.23	24.72	448.23	452.24	4.01
TabNet	367.37	454.30	86.93	454.30	454.63	0.33
FT-Transformer	556.97	1,153.15	596.18	1,115.65	1,115.84	0.18
RealTabR	557.18	1,068.71	511.53	701.70	763.43	61.73

Note. RSS = process-tree resident set size in MB, measured in separate processes at approximately 2-ms intervals. This table reproduces configuration-level resource measurements from the post-test replication, not memory readings of all original fitted instances.

All five classical models kept under about 282 MB in terms of their peak training RSS: Logistic Regression - 192.60 MB, Random Forest – 219.15 MB, XGBoost – 220.53 MB, LightGBM – 281.52 MB, and CatBoost – 280.08 MB. MLP got 448.23 MB and TabNet 454.30 MB. FT-Transformer and RealTabR demonstrated a significantly higher value for their training peaks – 1,153.15 MB and 1,068.71 MB, correspondingly. Incremental training peaks for them were also relatively big – 596.18 MB and 511.53 MB, and that is why the reason was not merely that the process baseline was high at the beginning.

In terms of inference, FT-Transformer demonstrated the highest peak RSS of 1,115.84 MB. RealTabR reached 763.43 MB. Peak RSS for XGBoost during inference was

219.16 MB, and its incremental increase was just 0.19 MB. Such low incremental numbers should not be considered as full memory requirements to serve a model; they do not take into account the process baseline and are specific to the stage of instrumented execution.

A necessary condition of fidelity is related to Table 12. The computation with fixed hyperparameters yielded the exact same probabilities as the original ones for Logistic Regression and MLP but not for all estimators supported by libraries. The memory values thus obtained may still serve as rough approximations of footprints for the designated configurations, but cannot be treated as direct measurements of memory usage for the original fitted objects. No reproduced prediction substituted an unchangeable output, and no memory footprint obtained after testing was used in tuning or classification.

Non-serializability of the final RealTabR estimator was a separate problem from the operational point of view. Locked values of predictions and timings for the estimator were preserved, while the equivalent final object for deployment was not obtained due to the use of local lambda.

4.9.3 Performance-efficiency synthesis

Figure 9 compares locked ROC-AUC to fit time on a logarithmic time scale. Such a comparison highlights the excellent performance-efficacy ratio of gradient boosting while recognizing that speed is not a metric of scientific quality standing alone. Logit Regression took less time to train than XGBoost, but its significant discrimination disadvantage rendered it a poor benchmark to compare against for churn risk ordering. MLP had an extremely small inference time measurement, and despite its larger training overhead, no discrimination edge could be gained over the other boosting methods.

XGBoost is a good compromise within the tested environment, since it achieved the highest PR-AUC, lowest Brier score, almost the highest ROC-AUC, cheap training cost, small model size, and relatively good resistance to masking and input stress. CatBoost's better ROC-AUC, Random Forest's better F1-score, RealTabR's better MCC, and MLP's smaller inference time prove that 'best model' is always a conditional statement.

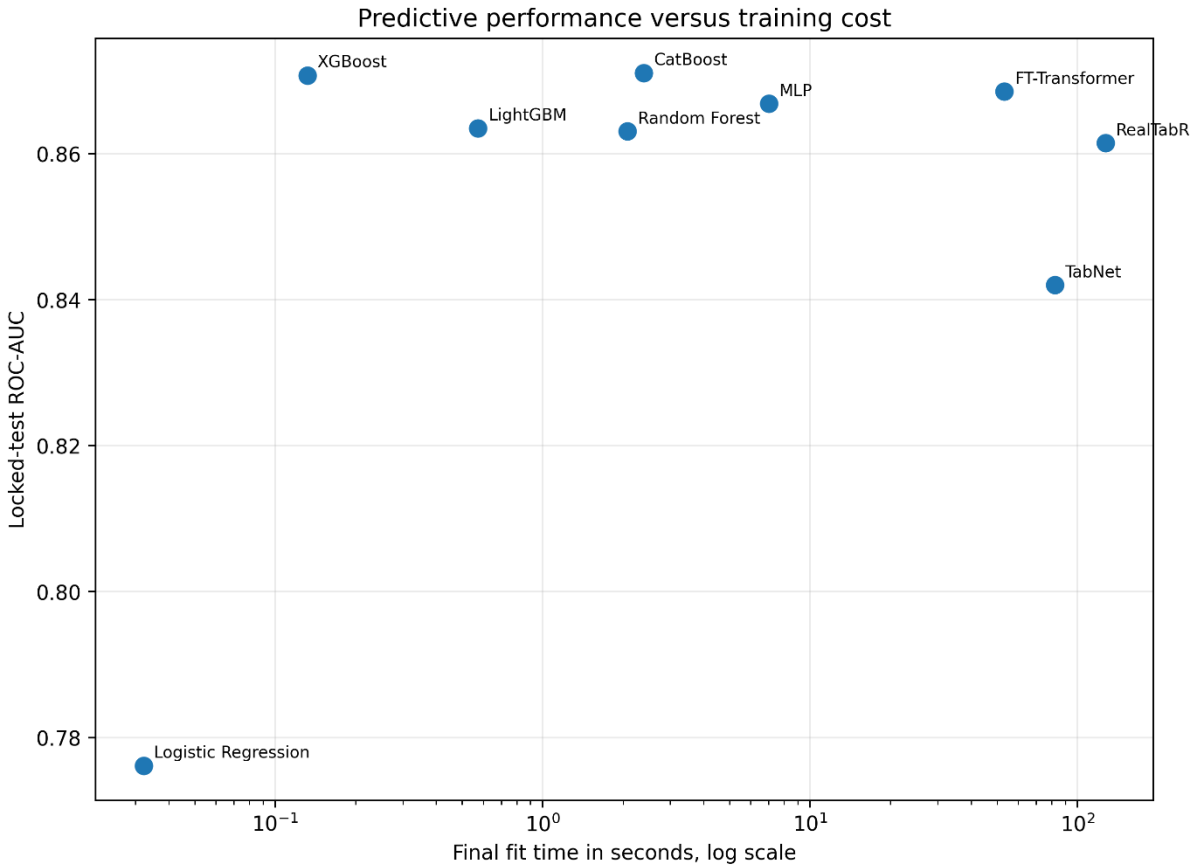


Figure 9. Locked ROC-AUC Versus Final Model Fitting Time

As a whole, the results obtained by computation reinforce the central result that complexity in the model does not necessarily yield a clear-cut benefit when the sample is of medium size and includes ten features in the tabular data format. The results also form the basis of selecting a particular model for the associated prototype mentioned in Chapter 5. Such a selection will always be a pragmatic one, based on the benchmark sample.

4.10 Chapter Summary

Tuned gradient-boosting approaches were shown to still perform very competitively against deep tabular models even under the leakage-controlled setting. CatBoost gave the best locked ROC-AUC (0.8711), while XGBoost showed the best PR-AUC (0.7223) and lowest Brier score (0.0986). Pairwise and multiplicity adjusted hypothesis testing found no differences in ROC-AUC between XGBoost and CatBoost, FT-Transformer, or MLP, but found multiple reliable differences against weak baselines. This resolves the comparison and uncertainty aspects of RQ1 and RQ2 without resorting to performance rankings of models.

Permutation SHAP analysis across models identified Age, NumOfProducts, and IsActiveMember as leading predictors in all cases, explaining about 71% of normalized four-model importance. The additional TabNet analysis confirmed the leading three predictors with prediction fidelity, while RealTabR could not be included in such SHAP

analysis due to failure to satisfy the pre-defined fidelity tolerance for the final model reconstruction. Robustness testing identified simulated input masking as a more harmful attack type than either noise addition or categorical substitutions. Subgroup analysis found substantial variations in base rates, error behaviors, and direct probabilities. Computational evidence favored a performance-efficiency compromise rather than a general advantage for larger deep architectures.

The conclusions of all this chapter are only dependent on the publicly available data set, the locked development and hold out process, and the post hoc diagnostics criteria provided. The implications of all these findings are highlighted in chapter 5 to explain how the selection of the XGBoost model can be used to prioritize the portfolios in a research environment.

CHAPTER 5. DISCUSSION AND PRACTICAL IMPLICATIONS

The performance and diagnostic metrics of nine bank churn models were analyzed in Chapter 4 using a consistent assessment framework. Chapter 5 reviews these findings and explores the potential to use a selected model as part of a bounded decision-making process. Three questions are delineated in this regard: what is shown in the data based on the predictions, which of the models is a justifiable selection for this particular application, and what other applications cannot be justified using the presented data.

5.1 Integrated Discussion of Research Findings

5.1.1 Predictive performance in relation to model complexity

The key comparative finding is that higher complexity in the neural models has not provided any consistent predictive superiority on the tested banking data set. The CatBoost model had the highest locked test ROC-AUC at 0.8711, whereas the XGBoost model demonstrated the highest PR-AUC (0.7223) and the lowest Brier score at 0.0986. The FT-Transformer model showed the highest ROC-AUC among the deep learning models at 0.8685, followed by MLP (0.8669). Those differences are negligible compared to the range between the best models and the linear Logistic Regression model. Therefore, the appropriate conclusion should be not about the ineffectiveness of the neural models, but rather that they have not succeeded in displacing the competitive gradient-boosted trees under the fixed research conditions.

Paired inference helps to draw more specific conclusions. With Holm correction applied, the ROC-AUC difference between the XGBoost and the other three algorithms (CatBoost, FT-Transformer, and MLP) has not been statistically significant. However, it does not prove the equality or equivalence of those models. It means that the holdout sample has not provided enough evidence of a difference for those specific model pairs. The statistical result matters because ordering models by the fourth decimal place could otherwise create a misleading sense of certainty about a winner.

The orderings of models differed depending on the objective too. Random Forest was the best in the locked F1-score and balance, whereas RealTabR achieved the maximum MCC value while employing a more conservative classification approach. A system that favors a fixed number of contacts encounters a different challenge from one employing a locked binary threshold. Discrimination measures ranking quality; Brier score measures probability quality; F1 and MCC measure particular threshold selection. These metrics serve different purposes, thus they cannot be used interchangeably when choosing a model (Saito & Rehmsmeier, 2015; Niculescu-Mizil & Caruana, 2005).

Data context matters. Ten thousand records with ten static features provide enough data to experiment with nonlinear tabular learning, but lack time-series transactions, history of account events, or interaction with customers. Transformer representation learning can prove to be more beneficial in a richer data regime, but this hypothesis cannot be proved on the current benchmark. The observed competitiveness of boosting is consistent with larger tabular-learning studies that also report strong tree-based

performance on many medium-sized structured datasets (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022).

5.1.2 Why validation, explanation, and robustness change the interpretation

The evaluation protocol is an integral component of the results. Hyperparameters, class imbalance handling, calibration, threshold setting, and early-stopping criteria were determined based on the 8,000 customers of the development dataset before the 2,000 customers of the sealed holdout set were evaluated. Bootstrap analysis, statistical analysis, explainability, and perturbation analysis did not change the model predictions. This protocol is essential to ensure that observations about the model's performance on the holdout data cannot influence further development of the model and thus introduce optimistic biases in benchmarking (Kapoor & Narayanan, 2023).

The lack of a predictive edge of the deep learning models becomes much more revealing once explanation stability and computational expenses are taken into account. Cross-model Permutation SHAP ranked the variables Age, NumOfProducts, and IsActiveMember first through third in the list of the most significant in XGBoost, CatBoost, MLP, and FT-Transformer, respectively. These features explained about 71% of mean normalized importance in the four models. TabNet extension maintained the top three set. However, at the same time, the inability to reproduce the RealTabR estimator used for SHAP explains that the explanation must be verified, not assumed. Model agreement strengthens the descriptive interpretation of fitted predictors; it does not turn their learned associations into causal explanations (Lundberg & Lee, 2017; Aas et al., 2021).

Robustness analysis and subgroup analysis show more limitations. Feature masking resulted in more damage than numeric noise and category switching, and different age groups, genders, geographic locations, and activity groups had different rates of churn and errors. Direct sensitivity analysis showed that there were changes in predictions when the input values of Gender and Geography were changed while other inputs were unchanged. This analysis does not prove discrimination. The role of these analyses is to find out behaviors which need further examination before using the model by an organization (Hardt et al., 2016; Pleiss et al., 2017).

5.1.3 Relationship to prior banking churn research

The banking literature provides useful context, but published numerical results cannot be placed on a single leaderboard without comparable data and protocols. de Lima Lemos et al. (2022) investigated a large Brazilian institutional dataset and conventional machine-learning alternatives. Brito et al. (2024) considered a much larger transaction-rich retail-banking environment, where feature engineering and imbalance procedures were central. Singh et al. (2024) studied a related public banking dataset using machine-learning models and a visualization application. Văduva et al. (2024) emphasized probability calibration and SHAP for two tree-ensemble models on a separate synthetic banking dataset. Their input information, churn rates, outcome definitions, and validation designs differ from those used here.

Table 13. Methodological Comparison with Representative Banking Churn Studies

Study	Data and scope	Analytical emphasis	Key difference in interpretation
de Lima Lemos et al. (2022)	500,000 customers from a Brazilian financial institution	Conventional ML comparison with repeated validation	Richer institutional data; scores not directly comparable
Singh et al. (2024)	10,000-customer public banking dataset	Conventional ML, XGBoost, SMOTE comparisons and visualization app	Related data family; different features and preprocessing
Brito et al. (2024)	Over 3 million customers and approximately 170 million transactions	Feature engineering, imbalance treatment and XGBoost	Transactional setting and much lower churn prevalence
Văduva et al. (2024)	10,000 records from a synthetic banking dataset	Random Forest/LightGBM, probability calibration and SHAP	Different data-generating setting; narrower model family
Present research	10,000 records, ten predictors; sealed 2,000-customer holdout	Nine-model comparison, paired inference, cross-model SHAP, robustness and efficiency	Controlled within-dataset evidence; no independent external validation

Note. Studies differ in features, prevalence, sampling, and validation. This table compares methodological scope rather than ranking published scores.

Table 13 thus places the current effort according to its scope of methodology rather than any achievement of a novel high predictive score. The unique aspect of the current effort in the context of its application environment is the use of five classical models and four deep tabular models, development-only model selection, a closed-box final assessment, multiple-hypothesis paired comparisons, cross-explanation testing of models, input perturbation testing, subgroup analysis, and computational assessment. Such a range of coverage might help interpret even a small difference in score and justify even a sub-optimal model selection decision.

5.2 Model Selection for the Application Prototype

5.2.1 Selection based on performance-efficiency trade-offs

Selection of XGBoost as the single scoring model in the application was done due to the balance between prediction performance and resource consumption. It had the second highest locked ROC-AUC of 0.8707 compared to CatBoost but had the best PR-AUC and Brier score. Final training time measured in the study was approximately 0.132 s and the size of the saved model artifact was 0.276 MB. FT-Transformer had a comparable ROC-AUC but much higher fitting time and artifact. These figures refer to the CPU-only setup of the study and do not ensure the response times and operating expenses in a production environment.

Choosing XGBoost does not mean that CatBoost, MLP, or FT-Transformer is worse in all possible cases. CatBoost will be an equally strong rival if the ROC-AUC and native categorical variables support are of priority. Also, a bank might choose some other solution if the respective consequences of missing the churners, making unnecessary calls, and providing the treatment are different. The prototype makes one reasonable

research decision based on frozen data but does not solve the problem of model selection for the bank.

5.2.2 Application components and evidence boundaries

The demonstration is built on top of a Streamlit interface, where an XGBoost classifier is serialized, along with a function that transforms the input ten features explicitly, predictions, SHAP background, and archived studies' evidences. Four sections are available: Overview, Individual Prediction, Portfolio Prioritization, and Transparency & Governance. The model is loaded from an XGBoost native JSON file, and the feature transformation process makes sure that the necessary columns exist and generates the correct numerical and one-hot encoded features. The saved threshold of the application is 0.392 and corresponds to the selected one during development.

The implementation caters both to individual and portfolio use case scenarios. The individual section loads the customer's profile and calculates the new prediction, whereas the portfolio section either scores an uploaded CSV file or applies the provided locked holdout as the example portfolio. The difference between the two sections and the governance section is that the latter provides a pre-existing summary of the study that has been done. In other words, the interface does not run nine different models, SHAP consensus bootstrap procedure, perturbation experiment, and sub-group statistics every time the user enters it.

The whole work, including the demonstration code and all the necessary artifacts, is described in https://github.com/MinhNguyen2323/bank-customer-churn-xai. This is just a reproducibility guide and not proof that the system has been deployed or that the model has been validated by someone else. The images referred to below are examples of the interfaces but the metrics remain those in Chapter 4.

5.3 Individual Customer Churn Prediction

5.3.1 Overview dashboard and interpretation of displayed metrics

Overview of the application is presented in Figure 10. The four summary figures for it include: locked XGBoost ROC-AUC (0.8707), PR-AUC (0.7223), Brier score (0.0986) and decision threshold (0.392). Visualization of both the discrimination power and the probability quality prevents from using the simple accuracy figure interpretation. The left pane is the distribution of the predicted probabilities on the locked 2,000 customers' holdout set with locked threshold specified; the right pane is the global SHAP importance for the demonstration XGBoost model. The plots describe the sample used for research and the fitted model, not the risk profile of an existing bank.

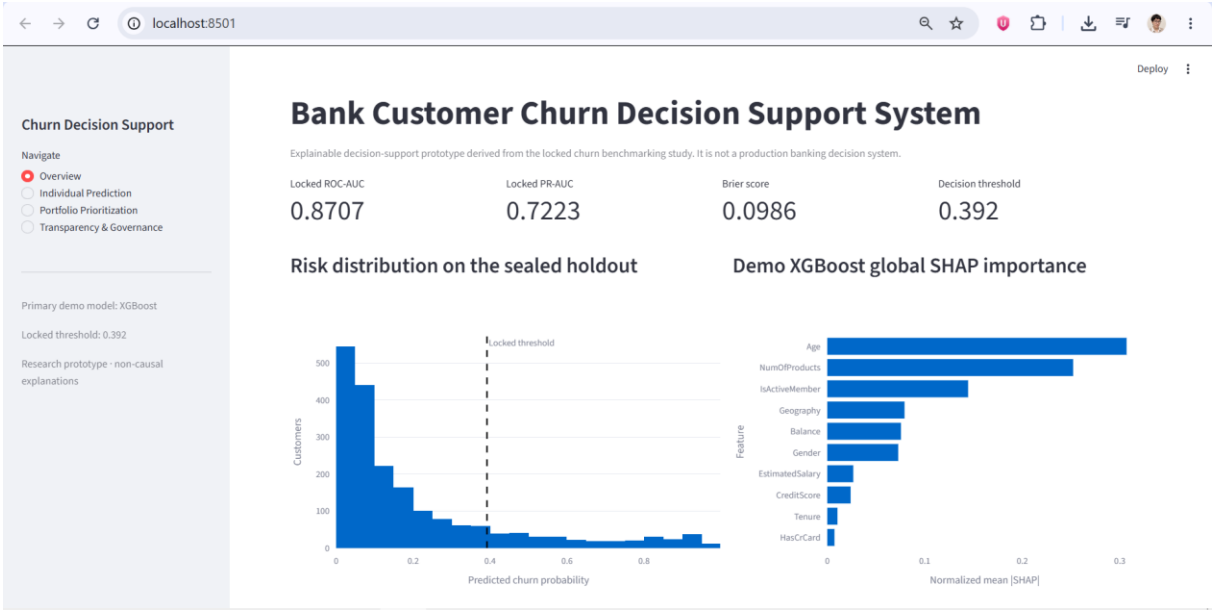


Figure 10. Application Overview Dashboard

Although the SHAP chart overview in Chapter 4 and SHAP consensus across models have similar yet distinct functions, the difference between them is that, for the case study artifact, contributions in probability scale for the intervention background in XGBoost were computed with the help of the TreeExplainer. The feature contribution aggregation is presented in the form of the normalized global importance. On the contrary, the research consensus used model-independent Permutation SHAP on four representative models with their common protocol. Comparison between the most prominent features of both models is possible on the general agreement level only.

5.3.2 Customer input interface and predictor consistency

Figure 11 illustrates the input screen for a single customer. It has ten fields which directly map to the predictors used across the experiment: CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. CreditScore, Age, Tenure, Balance, NumOfProducts, and EstimatedSalary are typed in as numeric values, whereas the two categorical and two status values are chosen from the available options. The form does not ask for any complaints, satisfaction, transactional details, or loyalty details which are not present in the training model.

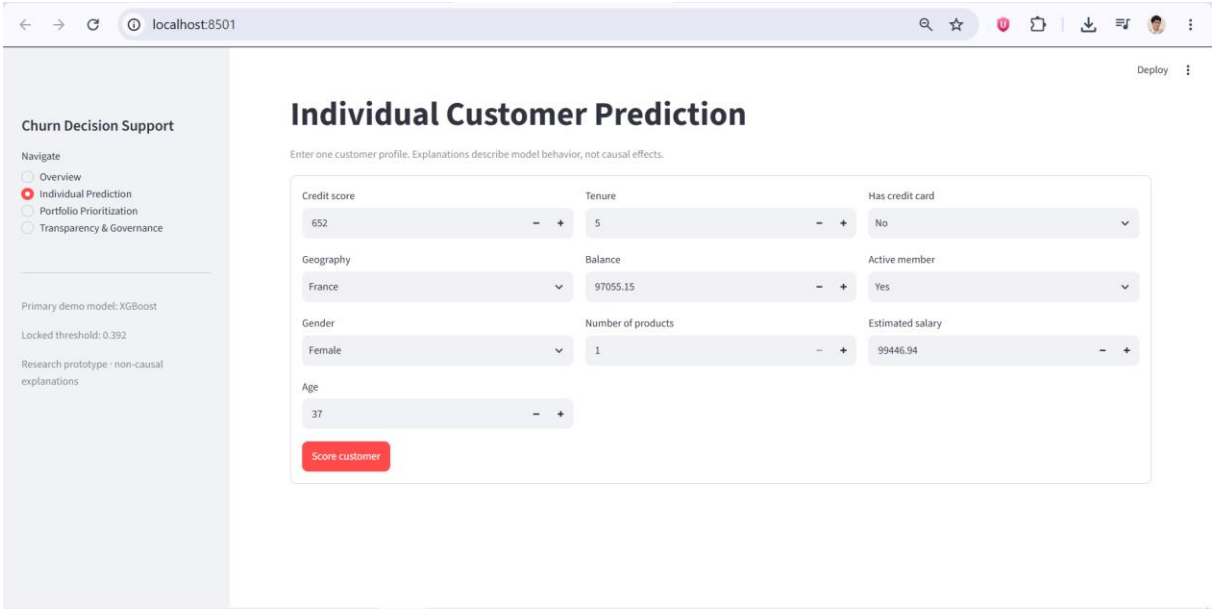


Figure 11. Individual Customer Input Interface

Input Value ranges and default values in the form are obtained from metadata from the development set. This makes the demo easy to use and helps to prevent out-of-range entries, but these should not be seen as replacements for the specific data validation at the bank. The uploaded portfolios are checked for mandatory fields and allowed categories; any errors in numeric conversions are not quietly converted into new model inputs. The target variable "Exited" is not a model input used for prediction. This distinction of input from output preserves the meaning of the prediction process.

5.3.3 Probabilities, descriptive risk tiers, and local explanations

Following submission of the customer's profile, the output interface shows the probability of churn, the risk category of the prediction, a fixed decision, and an explanation plot of SHAP contributions in the locality of the prediction (see Figure 12). The cutoff point of 0.392 makes the decision of whether the customer is "flagged for review" or "not flagged." The user interface risk categories are descriptive visualization categories where a "Low" value is below 0.20, "Moderate" is between 0.20 to 0.392, "High" is between 0.392 to 0.70, and "Critical" is above 0.70. The additional risk categories were not separately chosen or validated.

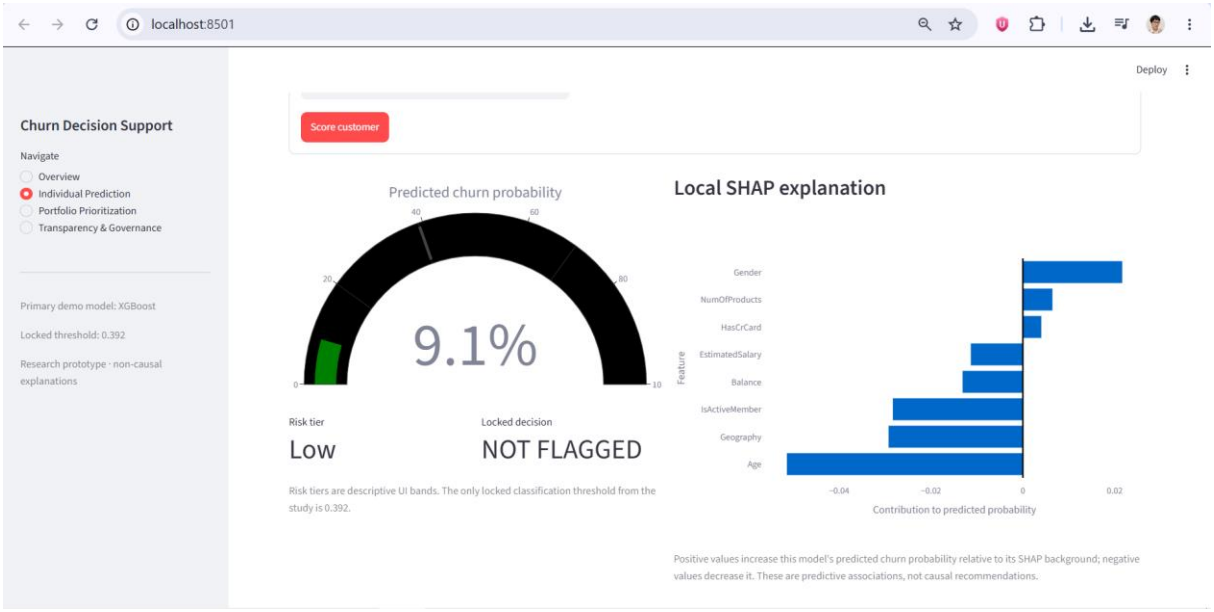


Figure 12. Individual Prediction and Local SHAP Explanation

Local Explanation – This breaks down the probability of a specific model relative to the background expectation of the explainer. SHAP values that are positive mean those features which increased the probability of this case relative to the reference and those which decreased it are SHAP negative. For the illustration, the one-hot encoding parts of Geography and Gender are reverted to their original input variable counterparts prior to displaying. It aligns better with the ten-input feature format but SHAP is definitely not an estimate of what would happen to churn if the bank intervened on the feature.

For instance, a customer whose age or activity status made a major contribution to the prediction could be looked into further but the explanation doesn't provide an offer for retaining him or her. Also, a change to an immutable feature through a hypothetical interface computation can't be considered an intervention. The model predictions help the analyst formulate his or her questions but treatment decision needs more causal knowledge, understanding of the customers, and other precautions.

5.4 Portfolio-Level Customer Prioritization

5.4.1 Capacity-constrained selection as a decision-support use case

An important use of churn ranking occurs when the retention effort is unable to reach all customers. The portfolio page ranks predicted probabilities by descending order, computes a priority rank, and enables the user to define a review capacity as a fraction of the size of the portfolio. The included software supports a capacity of 5% to 50%, in five percent intervals. The program picks out the first ceiling(capacity × number of customers) entries on the sorted list. This approach is a deterministic resource allocation decision for review and not a learned retention policy nor economic optimization of offers.

The demonstration included with this package uses the identical 2,000 customer sealed holdout used for reporting the performance of predictions. For a capacity of 20%, the

software chooses the 400 highest score customers. Out of 407 actual churners in the holdout, 258 fall in the chosen group, representing 63.4% recall at the top 20%. Figure 13 shows this chosen number and evaluation only capture results. The figure gives a concrete interpretation of a ranking metric: an analyst reviewing one-fifth of the records encounters almost two-thirds of the observed churn outcomes in the research dataset. These are historical label-based evaluation counts, not actual customers retained.

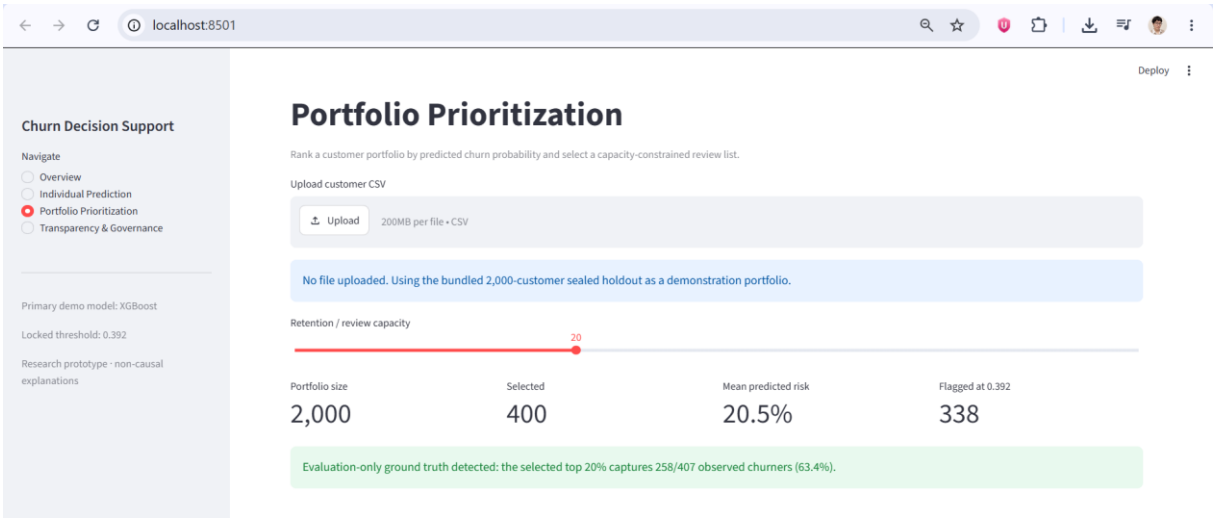


Figure 13. Portfolio Prioritization Summary at 20% Capacity

Only if the provided portfolio has the "Exited" label as the actual outcome will the capture metric be shown. The prototype issues a warning that such a metric will make sense only on records that have not been used before in training and should not be considered an independent test if such records were used earlier. The held out data set is the original sealed data set and not any new one. Any reuse of it for a demo will not constitute independent evidence of predictivity or success of the actual retention team.

5.4.2 Separating ranking capacity from the locked decision threshold

The result generated from the process has been shown in Figure 14 above where customers have been listed in ascending order according to their probability of churning. The list comprises of the following information: Priority Rank, Customer Id, Churn Probability, Risk Tier, Locked Decision, and Selected for Review. The CSV output ensures that all the scored variables remain in the list so that it can be viewed outside the platform. The main functionality of the prototype is to identify risky customers for manual review.

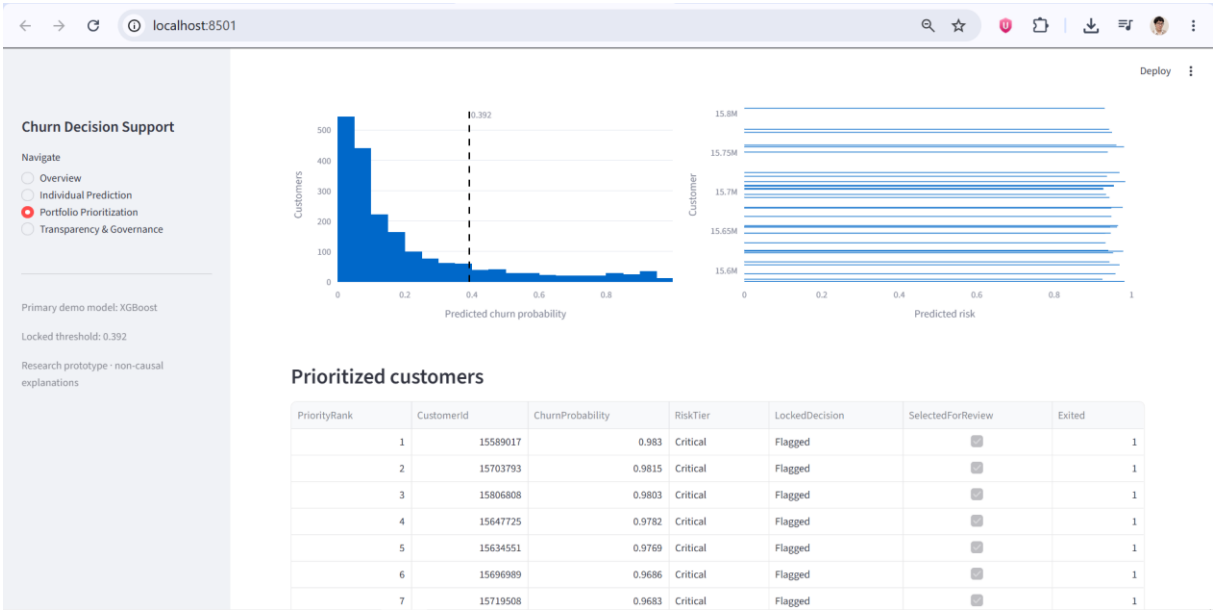


Figure 14. Prioritized Customer List

Locked classification threshold and capacity selection are two different notions. In the case of bundled holdout, out of 2,000 clients, 338 reach probability of ≥0.392 and become classified as Flagged. However, at 20% capacity, 400 clients are selected through ranking, and out of them 62 fall below that classification threshold. Therefore, SelectedForReview solves the capacity problem, while LockedDecision solves the predefined binary classification problem. Disagreement between them on some records is possible and normal; no adjustment is to be done based on holdout results after locking the evaluation.

The same distinction makes impossible to make inflated claim about operational impact of selected clients. Selecting 400 clients and determining 258 as churned observed does not mean that 258 departures could be avoided. Effectiveness of retention depends on the effectiveness of the action, reaction of customers on it, costs of conducting such action and opportunity cost of contacting the client who would not churn anyway. No such information is available in the provided dataset. Therefore, the legitimate application result is improved concentration of observed churn risk under a capacity constraint, not measured financial or causal retention benefit.

5.5 Explainability and Governance Integration

5.5.1 Cross-model explanation evidence alongside the selected model

The single-model dashboard may create a tendency in the user to take a feature ranking of one particular estimator as an explanation for the client's behavior. It is solved with the help of the governance page, which shows the normalized Permutation-SHAP consensus of four models according to the benchmark (Figure 15). The features Age, NumOfProducts, and IsActiveMember make up about 71% of the mean normalized importance in XGBoost, CatBoost, MLP, and FT-Transformer models. The same three features were identified as important with the help of the additional analysis of TabNet.

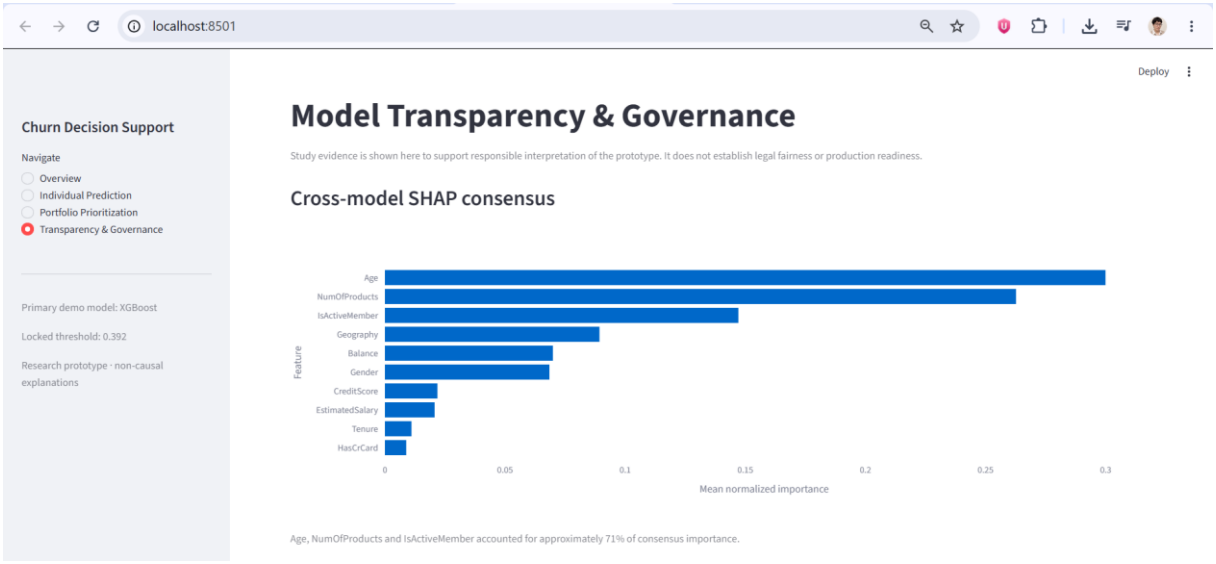


Figure 15. Cross-Model Explainability and Governance

Such stability of rankings across the globe is helpful, though it does not nullify the interpretation issues. The number of products had a non-linear relationship, rather than a linear rule stating that the increase in the number of products will always result in decreased risk. The group of three-product customers in the explanation sample using SHAP method was also small, limiting any interpretation at that level. The feature correlation, distribution, and attribution algorithm used can also affect SHAP values. This is why the governance perspective should express consensus and disagreement, without turning predictive explanations into controllable drivers (Aas et al., 2021).

5.5.2 Robustness, subgroup, and direct-sensitivity diagnostics

Figure 16 summarizes three diagnostic visualizations that complement each other. First, the page on governance includes certain estimated changes to the approximate XGBoost ROC-AUC scores due to deliberate perturbations: roughly –0.0064 under moderate numeric noise, –0.0242 under 10% masking, –0.0050 under 10% categorical substitution, and –0.0200 under combined stresses. They are not new stress test results but the summaries of the fixed study stresses performed previously, and the more pronounced impact of masking justifies the focus on the input completeness and input value unavailability handling methods.

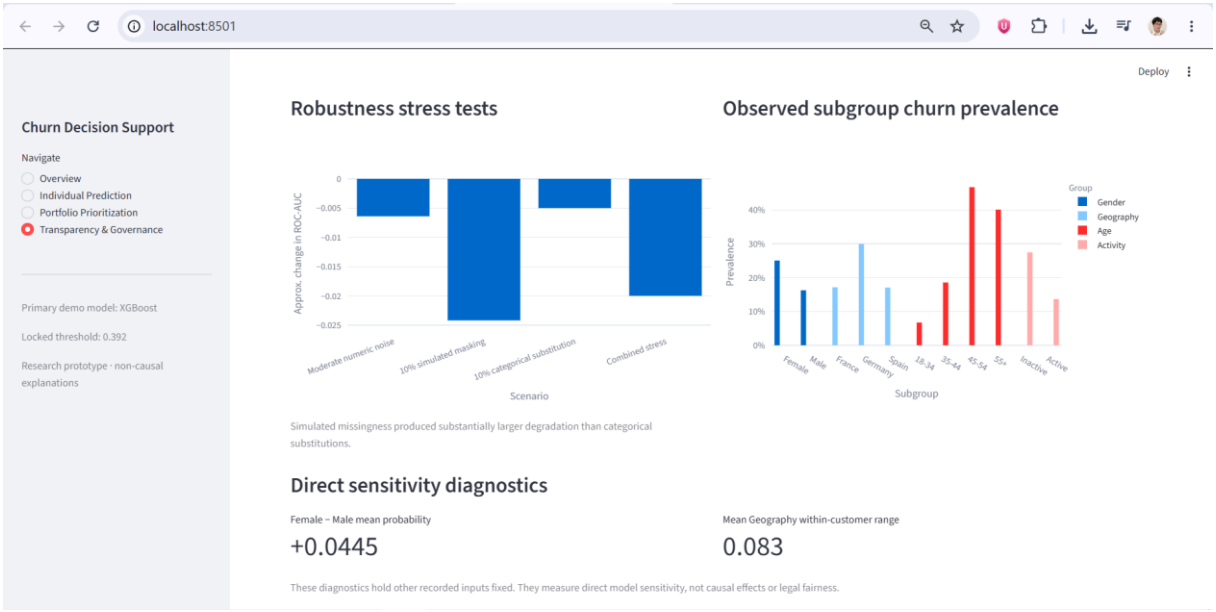


Figure 16. Robustness, Subgroup, and Sensitivity Governance

Second, the interface provides the prevalence of churn observations by Gender, Geography, age cohorts, and activity status. As an example, the prevalence of the archived holdout is 25.05% for Female and 16.26% for Male clients; Germany has 29.96% as compared to 17% in France and Spain. Base rate differences are very important background information in assessing alarm rates and errors of the groups under analysis, yet only prevalence differences cannot prove the existence of any bias and causality of the predicted association between the variables.

Third, direct sensitivity is provided for Gender and Geography. For the static XGBoost algorithm, the mean difference in probabilities of assigning Female instead of Male for the same profiles is approximately 0.0445; the mean within-client probability range in the Geography assignments is 0.083. Such calculations do not consider any variation in other recorded features and can produce non-representative of real clients' combinations of inputs. Therefore, such indicators reflect the response of the model to manipulation of certain inputs rather than causation and certification of fairness. Decisions involving sensitive attributes would require review of institutional policy, applicable law, alternative model specifications, and independently validated subgroup behavior.

In addition, the governance interface includes limitations in form of a single public dataset, static predictor variables, lack of any interventions' outcomes, and lack of independent institutional or temporal assessment. The significance of the governance interface is conceptual and not regulatory. What it demonstrates is the extent to which an analyst-oriented prototype tool can highlight key uncertainties and model dependencies. It does not create accountability practices like access authorization, approval of model changes, logging, and so forth.

5.6 Practical Implications and Research Limitations

5.6.1 Implications for a cautious bank-retention workflow

In combination, these results suggest a bounded workflow where risk scoring is used to prioritize review and not automate treatment. In practice, an organization might generate lists of candidate clients using existing client features, investigate local reasons for strange predictions in the region of interest, and allocate scarce resources for review based on their rankings. Data integrity and business context would have to be confirmed manually before a final decision could be made to contact a client. This is an example of a possible workflow suggested by the demonstration, not an implementation and evaluation of an operational policy of a particular bank.

A first takeaway from this experiment is that probability score is an input into a decision-making process and not a decision per se. While a risk estimate that has been calibrated is useful for prioritization and scenario analysis, a decision about optimal action needs an estimate of treatment benefit, customer value, costs of contact and risks associated with false-positive prediction. Churn probability and preventable churn are different things. The future decision policy might be quite different from a constant MCC criterion used for the classification task or a 20% ranking demonstration capacity. Such choices should be specified with business and ethical objectives before operational testing.

Another contribution made by the reproducible artifacts is that the stored predictor schema, the frozen models' configurations, the threshold, the evaluation results, and the implementation documentation make it possible to trace the presented outcomes of the prototype back to the research design. By reproducing results here is meant only that it is possible to check the inputs and outputs saved from the application and use the stored artifact for scoring. This does not mean that all the estimators estimated during fitting were completely reproducible. It was stated for RealTabR that such complete reproducibility could fail even with the saved settings and split indices.

5.6.2 Threats to validity and boundaries of practical use

There are three factors in particular that should be considered when interpreting the prototype. First, external validity is not tested; all of the key predictions in the prototype have been made using one open-source dataset and one final stratified holdout. Development validation and paired holdout inference strengthen the internal validity but do not show how the model works for other institutions and at different times. Second, the predictor variables do not have any dynamic histories of transactions, interactions, product adoption and retention contacts. An actual bank will see completely different relationships among the predictors or lack some of the ten predictor variables recorded.

Third, robustness, subgroup and direct sensitivity analyses are diagnostics based on defined synthetic circumstances, and not a complete check of safety and fairness for operation. Computational times were obtained under a four-core CPU-only experimental setup. Retrospective memory values are static values, not maximum memory recorded in the original fitting process. The Streamlit demo has not undergone a target-institution evaluation of concurrent users, access control, secure data handling, monitoring, or production service reliability. Its current state does not justify a production-readiness claim.

Another differentiation is between outcomes observed and effects of intervention. The Exited classification in the holdout data is the observation of churn in the benchmark sample, while there is no observation of customers assigned randomly to contact or not, no uplift calculation, and no observation of retention following contact in the application. Neither does risk concentration nor SHAP attribution prove that any specific customer can be convinced to stay.

5.6.3 Synthesis and transition to the conclusion

Combined, the findings suggest that the best possible contributions are neither a novel neural network design nor a ready-to-use churn prediction solution. They lie in a well-defined way from the controlled comparison of models to the informed selection of models based on evidence and the creation of an easily inspected research prototype. XGBoost is the right performance-efficient approach to apply to this data set; cross-model SHAP confirms the consistency of top predictive correlations; while the robustness, subgroup, and sensitivity analysis detects the behaviors that should be investigated further. The application uses this information to generate individual scores and sorted reviews without asserting any causality in terms of customer retention. Chapter 6 presents the final conclusions regarding the four research questions and the necessary future work.

CHAPTER 6. CONCLUSION AND FUTURE RESEARCH

6.1 Summary of Principal Findings

In this thesis, the reliability of the modern deep tabular learning compared to conventional machine-learning techniques was tested as far as predicting the bank customer churn rate is concerned. Instead of focusing on the performance of one metric, this research combined nine-model benchmarking with leakage prevention, probability quality, paired hypothesis testing, explanation stability, robustness, subgroup properties, and computation. These conclusions are unique to the public data set of 10,000 customers and the locked evaluation method outlined in Chapters 3 and 4. The four research questions can be answered as follows.

6.1.1 RQ1: Comparative predictive performance

No clear predictive superiority of the deep models over gradient-boosted trees has been proven based on the analysis of the results on the studied data. On the holdout sample of 2,000 clients, CatBoost demonstrated the best ROC-AUC (0.8711), while XGBoost had the best PR-AUC (0.7223) and the smallest Brier score (0.0986). The FT-Transformer performed best according to ROC-AUC among the deep models (0.8685), followed by MLP (0.8669). The locked thresholds for different measures provided different best models for F1-score, balanced accuracy, and MCC, indicating that discrimination, probability calibration, and binary classification performance are answering different questions.

6.1.2 RQ2: Reliability of model differences and development choices

Rankings should be qualified in a statistical manner. With post-Holm correction applied to the paired DeLong test, no statistically significant ROC-AUC performance difference was found between the XGBoost and CatBoost, FT-Transformer, and MLP. It means that we cannot prove the lack of differences, just that we have no evidence about such an existence. On the other hand, XGBoost performed better than the LightGBM, TabNet, and Logistic Regression regarding this metric under the corrected testing framework. Model ablation on development data only did not show superiority of the class weights and SMOTENC over the unaltered class distributions in accordance with the chosen set of multi-metric criteria. The MLP ablation showed the advantages of the dropout without Batch Normalization. Model-specific calibration and MCC thresholding were chosen beforehand without changes in view of the holdout performance hit of RealTabR's locked isotonic calibrator.

6.1.3 RQ3: Consistency and limits of explanation

The Permutation-SHAP analysis of XGBoost, CatBoost, MLP, and FT-Transformer determined the same three main predictors – Age, NumOfProducts, and IsActiveMember. They explained about 71% of the mean normalized importance of all models. The agreement in the leading associations was corroborated by cross-model rank correlation and bootstrapped stability. Moreover, the TabNet interpretation had the same three features as the most important but switched their first two positions. Age and

the feature that characterized the customers' activity level were rather stable in their directions of influence. However, the association between the number of products purchased and the prediction of churn is nonlinear. These are interpretations of the predictions but not the proof that changing the variables would cause any changes in behavior of the customers (Lundberg & Lee, 2017). The original RealTabR estimator cannot be reproduced anymore, and hence its SHAP values are excluded.

6.1.4 RQ4: Robustness, subgroup behavior, and computational cost

Controlled perturbations showed that the main models demonstrated more robustness to small numeric and categorical alterations rather than to the simulation of input value losses; FT-Transformer proved to be quite sensitive to numeric noise. Subgroup analysis uncovered significant discrepancies between prevalence and performance in terms of errors in respect of age, gender, geographic location, and activity status. Altering Gender or Geography while fixing other recorded characteristics affected certain model predictions too. These diagnostics reveal model dependencies that should be analyzed, but not serve as evidence of fair treatment or causality. Computer-generated performance measures additionally affected the interpretation of similar performance values: XGBoost demonstrated balanced discrimination and probabilistic performance along with the small cost of training, storing, and explaining in the four-core CPU-only environment.

6.2 Scientific and Practical Contributions

6.2.1 Contribution to empirical model assessment

Scientific Contribution

The key scientific contribution is a unified, auditable benchmark rather than an innovative machine learning framework. Five classical algorithms and four deep tabular architectures were compared under a shared ten-feature specification, a stratified development-holdout split, fold-local feature engineering, model-specific hyperparameter tuning, repeated development verification, and once-and-only-once evaluation on unchanging holdouts. Statistical testing of each pair of models helped to distinguish between observed ranks and significant differences. Together with calibration, explanation stability, perturbation robustness tests, subgroup analysis, and computational requirements, this added more information to the accuracy and ROC-AUC tables. The combined hyperparameter settings, extra attribution analysis for TabNet, RealTabR fidelity testing, and post-hoc memory usage analysis document both procedural and empirical reproducibility limitations.

6.2.2 Contribution to bounded decision support

The provided application converts the chosen XGBoost model artifact to churn probabilities per customer, local explanations using SHAP value approach, risk bands based description, and a constrained capacity review list. The application utilizes the same ten predictive variables and fixed classification threshold of 0.392, without mixing up the concepts of threshold-based classification and score-based selection. For the

provided 2,000 customers holdout sample demonstration, 20% capacity for review process resulted in choosing the 400 top customers out of which 258 out of 407 observed churners (63.4%) were included. Governance views display cross-model explanation evidence and archived diagnostic findings without presenting them as fresh model evaluations. The project repository makes the research implementation and demonstration artifacts accessible: https://github.com/MinhNguyen2323/bank-customer-churn-xai.

Collectively, these contributions provide a clear path from assessment to model selection and finally to a transparent decision support system prototype. These are not external validations by banks, optimizations of retention policies, certifications for fairness, or actual deployments into production systems. Maintaining this separation is key to understanding the thesis.

6.3 Future Research Directions

6.3.1 Independent and temporal validation

The primary objective should be to evaluate the frozen model using independent data from institutions that have a consistent definition of churn and customer properties. This will help to determine if the rankings, calibration, explainability, and subgroups are consistent away from the benchmark. If there are time-stamped data, testing using training in earlier time periods and validation in later periods could reveal generalizability, drift, and change in feature availability. Repeated cross-validation or bootstrapping of the same holdout sample during development cannot replace these evaluations.

6.3.2 Richer data, model governance, and robustness

Future research may consider transaction histories, digital activity trajectories, customer service, product usage, and previous retention interactions. The ability to do so would enable the investigation of whether richer representation learning models add any further advantage over the static-feature case investigated here. Robustness analysis should consider natural missingness, data recording mistakes, feature failures, and demographic changes rather than artificial disruptions. Consideration of sensitive feature inclusion and exclusion, subgroup-specific model calibration and error rates, and any potential for fairness mitigation is required along with pre-defined criteria for evaluating such claims and consideration of the institutional and legal context (Hardt et al., 2016). Input sensitivity studies should not be the sole basis for claims of fairness.

6.3.3 Intervention value and operational evaluation

The next applied project needs to draw a distinction between the prediction of churn and the prediction of preventable churn. The combination of risk scores, customer value, costs of outreach, communication capacity, and expected responsiveness to treatment may help develop a specific cost-sensitive approach to customer retention. The evaluation of the effects of intervention, conducted using randomized or any other justifiable method, will allow us to know if the outreach influences customers' behavior.

Once the prototype is independently validated in its predictive function, it will be possible to examine its operational safety.

Conclusively, there is no evidence to believe that increased complexity leads to better churn prediction in the selected benchmarking data set. The main lesson drawn from this benchmark is related to methodology: reliable conclusions should be made on the basis of understanding the predictive power in conjunction with uncertainties, explanatory capability, robustness, behavior of subgroups and computational cost. In this case, the gradient boosting method demonstrates high competitiveness, and the XGBoost-based system shows the way in which well-substantiated predictions may help to prioritize reviews of clients.

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